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THESIS

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Digital Twin for eHealth: State of the Art and Conceptual Architecture

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Abstract

The digital transformation of healthcare has led to the emergence of connected medical devices, electronic health records, remote monitoring platforms, and Artificial Intelligence-based decision-support systems. Although these technologies generate large amounts of heterogeneous data, many eHealth systems remain fragmented, static, and weakly personalized. Digital Twin technology offers a structured paradigm for addressing these limitations by connecting physical entities, processes, or patients with virtual representations that can be updated, analyzed, simulated, and used to support decisions.

This Master's thesis presents a structured state-of-the-art study of Digital Twins for eHealth. It reviews eHealth foundations and analyzes the evolution of Digital Twin definitions from the classical three-dimensional physical–virtual model toward extended data- and service-centered approaches. It then studies maturity levels and healthcare applications before developing two complementary states of the art. The first examines Digital Twins in eHealth and healthcare, including patient-specific twins, organ and device twins, hospital-process twins, public-health twins, Artificial Intelligence-enabled services, interoperability, privacy, ethics, governance, and validation. The second focuses on cardiovascular Digital Twins as a representative clinical deep-dive, covering electrophysiology, hemodynamics, heart-failure models, patient-specific calibration, hybrid AI–mechanistic approaches, population-scale cardiac twins, and validation challenges.

The thesis includes a comparison of representative Digital Twin definitions, two comparative landscape tables, and several academic synthesis figures that clarify definition streams, AI roles, eHealth Digital Twin categories, cardiovascular Digital Twin categories, maturity evidence, and the transition from the reviewed literature to the CardioTwin PFE. The work derives a conceptual multi-layer architecture and explicit design implications for CardioTwin while preserving the general eHealth scope of the Master's study. CardioTwin is introduced only as a practical cardiovascular continuation that will require implementation, evaluation, and future clinical validation.

Keywords: Digital Twin, eHealth, Digital Health, Healthcare, Cardiovascular Digital Twin, Artificial Intelligence, Simulation, Interoperability, Clinical Decision Support, CardioTwin.

Résumé

La transformation numérique du secteur de la santé a favorisé l'apparition de dispositifs médicaux connectés, de dossiers médicaux électroniques, de plateformes de télésurveillance et de systèmes d'aide à la décision fondés sur l'intelligence artificielle. Malgré la quantité importante de données générées, plusieurs systèmes eHealth restent fragmentés, statiques et faiblement personnalisés. La technologie du jumeau numérique offre une approche structurée pour répondre à ces limites en reliant des entités physiques, des processus ou des patients à des représentations virtuelles pouvant être mises à jour, analysées, simulées et utilisées pour soutenir la décision.

Ce mémoire de Master présente une revue structurée de l'état de l'art des jumeaux numériques appliqués à l'eHealth. Il étudie les fondements de l'eHealth et analyse l'évolution des définitions du jumeau numérique, depuis le modèle classique à trois dimensions reliant l'entité physique, l'entité virtuelle et la connexion, jusqu'aux approches étendues centrées sur les données et les services. Il étudie ensuite les niveaux de maturité et les applications dans le domaine de la santé, puis développe deux états de l'art complémentaires. Le premier concerne les jumeaux numériques en eHealth et en santé, notamment les jumeaux centrés sur le patient, les organes, les dispositifs médicaux, les processus hospitaliers, la santé publique, les services basés sur l'intelligence artificielle, l'interopérabilité, la confidentialité, l'éthique, la gouvernance et la validation. Le second se concentre sur les jumeaux numériques cardiovasculaires comme étude clinique représentative, en abordant l'électrophysiologie, l'hémodynamique, l'insuffisance cardiaque, la calibration patient-spécifique, les approches hybrides entre IA et modèles mécanistes, les jumeaux cardiaques à grande échelle et les défis de validation.

Le mémoire comprend une comparaison des définitions représentatives du jumeau numérique, deux tableaux comparatifs en orientation paysage et plusieurs figures de synthèse académique présentant les courants de définition, les rôles de l'IA, les catégories de jumeaux numériques en eHealth, les catégories cardiovasculaires, la maturité des preuves et la transition vers le PFE CardioTwin. Le travail déduit une architecture conceptuelle multicouche ainsi que des implications de conception pour CardioTwin, tout en conservant la portée générale du mémoire. CardioTwin est présenté comme une continuation pratique cardiovasculaire nécessitant une implémentation, une évaluation et une validation clinique futures.

Mots-clés : Jumeau numérique, eHealth, santé numérique, santé cardiovasculaire, intelligence artificielle, simulation, interopérabilité, aide à la décision clinique, CardioTwin.

الملخص

أدى التحول الرقمي في قطاع الصحة إلى ظهور الأجهزة الطبية المتصلة، والسجلات الصحية الإلكترونية، ومنصات المراقبة عن بعد، وأنظمة دعم القرار المعتمدة على الذكاء الاصطناعي. ورغم أن هذه التقنيات تنتج كميات كبيرة من البيانات المتنوعة، فإن العديد من أنظمة الصحة الإلكترونية ما تزال مجزأة، وثابتة، ومحدودة التخصيص. تقدم تقنية التوأم الرقمي إطارا منظما لمعالجة هذه الحدود من خلال ربط الكيانات أو العمليات أو المرضى في العالم الحقيقي بتمثيلات افتراضية قابلة للتحديث والتحليل والمحاكاة ودعم القرار.

يعرض هذا البحث مراجعة منظمة لحالة الفن حول التوائم الرقمية في مجال الصحة الإلكترونية. يتناول البحث أسس الصحة الإلكترونية ويحلل تطور تعريفات التوأم الرقمي من النموذج الكلاسيكي ثلاثي الأبعاد، القائم على الكيان الفيزيائي والكيان الافتراضي والاتصال بينهما، إلى النماذج الموسعة التي تجعل البيانات والخدمات أبعادا صريحة. ثم يدرس مستويات النضج والتطبيقات الصحية، ويطور دراستين متكاملتين لحالة الفن. تركز الدراسة الأولى على التوائم الرقمية في الصحة الإلكترونية والرعاية الصحية، بما في ذلك التوائم الخاصة بالمريض، وتوائم الأعضاء والأجهزة، وتوائم العمليات الاستشفائية، والصحة العامة، والخدمات المدعومة بالذكاء الاصطناعي، وقابلية التشغيل البيني، والخصوصية، والأخلاقيات، والحوكمة، والتحقق. أما الدراسة الثانية فتركز على التوائم الرقمية القلبية الوعائية كحالة سريرية ممثلة، وتشمل الفيزيولوجيا الكهربائية، والديناميكا الدموية، ونماذج قصور القلب، والمعايرة الخاصة بالمريض، والمقاربات الهجينة بين الذكاء الاصطناعي والنماذج الفيزيولوجية، والتوائم القلبية واسعة النطاق، وتحديات التحقق.

يتضمن البحث مقارنة للتعريفات التمثيلية للتوأم الرقمي، وجدولين مقارنين بوضعية أفقية، وعدة أشكال تركيبية أكاديمية توضح تيارات التعريف، وأدوار الذكاء الاصطناعي، وتصنيف التوائم الرقمية في الصحة الإلكترونية، والتصنيف القلبي الوعائي، ومستويات النضج والدليل العلمي، والانتقال من نتائج الدراسة إلى مشروع CardioTwin. كما يستخلص البحث معمارية مفاهيمية متعددة الطبقات وآثارا تصميمية لمشروع CardioTwin، مع الحفاظ على الطابع العام للدراسة حول التوائم الرقمية للصحة الإلكترونية. ويقدم CardioTwin كاستمرار تطبيقي في المجال القلبي الوعائي يحتاج إلى تنفيذ وتقييم وتحقق سريري مستقبلي.

الكلمات المفتاحية: التوأم الرقمي، الصحة الإلكترونية، الصحة الرقمية، الرعاية الصحية، التوأم الرقمي القلبي الوعائي، الذكاء الاصطناعي، المحاكاة، قابلية التشغيل البيني، دعم القرار السريري، CardioTwin.

ACRONYMS

AI	Artificial Intelligence
API	Application Programming Interface
BP	Blood Pressure
CDSS	Clinical Decision Support System
CPS	Cyber-Physical System
CVD	Cardiovascular Disease
DBP	Diastolic Blood Pressure
DL	Deep Learning
DT	Digital Twin
DT4H	Digital Twin for Health
eHealth	Electronic Health
EHR	Electronic Health Record
EMR	Electronic Medical Record
FHIR	Fast Healthcare Interoperability Resources
GDPR	General Data Protection Regulation
HL7	Health Level Seven
ICT	Information and Communication Technology
IoMT	Internet of Medical Things
IoT	Internet of Things
ML	Machine Learning
P4 Medicine	Predictive, Preventive, Personalized and Participatory Medicine
PFE	Projet de Fin d'Etudes
SaMD	Software as a Medical Device
SBP	Systolic Blood Pressure
TRL	Technology Readiness Level
AF	Atrial Fibrillation
CFD	Computational Fluid Dynamics
CMR	Cardiac Magnetic Resonance
ECG	Electrocardiogram
HF	Heart Failure
LLM	Large Language Model
MRI	Magnetic Resonance Imaging
RHC	Right Heart Catheterization
TTE	Transthoracic Echocardiography
VT	Ventricular Tachycardia

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Part I

General Introduction

CHAPTER 1

GENERAL INTRODUCTION

1.1 Context

Healthcare systems are increasingly supported by digital technologies such as connected medical devices, electronic health records, telemedicine platforms, mobile health applications, and Artificial Intelligence-based analytics. This evolution has contributed to the emergence of eHealth, a broad domain that aims to improve healthcare delivery, monitoring, prevention, diagnosis, and patient engagement through information and communication technologies.

At the same time, healthcare data are becoming more heterogeneous and continuous. A patient may generate information through clinical consultations, laboratory tests, medical imaging, wearable devices, mobile applications, and hospital information systems. The challenge is no longer only to collect data, but also to transform these data into meaningful, reliable, and actionable knowledge for healthcare professionals and patients.

Digital Twin technology has recently gained attention as a paradigm for representing physical entities and processes through dynamic virtual counterparts. However, the term is used with different meanings across product lifecycle management, aerospace, manufacturing, information systems, and healthcare. This thesis therefore analyzes the evolution of Digital Twin definitions from the classical three-dimensional model toward extended data- and service-centered models before applying the concept to eHealth. In healthcare, a Digital Twin may support the representation of a patient, an organ, a medical device, a hospital service, a clinical workflow, or a population-level process. The objective is to connect real-world data with virtual models in order to monitor current states, simulate possible futures, and support decision-making.

1.2 Problem Statement

Despite the progress of eHealth systems, several limitations remain. Medical data are often distributed across multiple systems and formats, which makes integration and interoperability difficult. Many digital health applications also provide static analysis based on isolated measurements rather than dynamic representations that evolve with the real-world situation. In addition, decision-support systems frequently rely on general population models and may not sufficiently account for individual variability, contextual information, uncertainty, and temporal evolution.

These limitations raise the following central problem:

> How can Digital Twin technology be used as a conceptual and architectural foundation for building eHealth systems that are more dynamic, interoperable, personalized, and capable of supporting monitoring, simulation, and decision-making?

1.3 Research Motivation

The motivation of this Master's thesis comes from the need to move from fragmented eHealth applications toward integrated and intelligent healthcare ecosystems. A Digital Twin-based approach can provide a structured way to combine real-world data, virtual representation, simulation, Artificial Intelligence, and decision services. This is particularly relevant in healthcare because clinical decisions depend not only on current measurements, but also on patient history, context, uncertainty, and possible future trajectories.

While the thesis examines Digital Twins for eHealth in a general healthcare context, cardiovascular Digital Twins are used as a representative clinical deep-dive. This domain combines multimodal biosignals, imaging, hemodynamic measurements, longitudinal monitoring, and patient-specific physiological modeling, making it suitable for analyzing the transition from conceptual healthcare Digital Twin architectures to clinically oriented implementations. This running case also establishes the theoretical basis for the cardiovascular specialization introduced later in the CardioTwin PFE.

1.4 Objectives

The main objective of this thesis is to study the state of the art of Digital Twin technology in eHealth and to identify how the literature can inform a practical cardiovascular PFE specialization. The specific objectives are:

- to review the foundations of eHealth, digital health systems, and Digital Twin concepts;
- to compare representative Digital Twin definitions and distinguish classical, extended, application-oriented, and configuration-oriented definition approaches;
- to distinguish digital models, digital shadows, connected Digital Twins, and intelligent Digital Twins;
- to analyze Digital Twins in healthcare through a general eHealth state of the art;
- to analyze cardiovascular Digital Twins as a representative clinical deep-dive;
- to compare selected studies through two landscape state-of-the-art tables;
- to synthesize architecture, data-flow, Artificial Intelligence, validation, and governance requirements;
- to derive literature-based design implications for the CardioTwin PFE.

1.5 Contributions

The main contributions of this thesis are:

- a structured synthesis of Digital Twin concepts and their relevance to eHealth;
- a comparative analysis of Digital Twin definitions, including the classical three-dimensional model, the extended five-dimensional model, and the operational definition adopted in this thesis;
- a dual state-of-the-art review covering both general healthcare Digital Twins and cardiovascular Digital Twins;
- a comparative analysis of representative eHealth and cardiovascular studies using two landscape tables;
- academic synthesis figures covering Digital Twin definitions, Artificial Intelligence roles, healthcare Digital Twin categories, cardiovascular Digital Twin categories, and maturity evidence;
- a discussion of interoperability, privacy, security, explainability, validation, and governance as cross-cutting requirements;
- a set of literature-derived design implications that connect the Master's thesis findings to the CardioTwin PFE.

1.6 Thesis Outline

This thesis is organized into five main parts. Part I introduces the research context, problem statement, objectives, and contributions. Part II first presents Digital Twin fundamentals, including the evolution of definitions, the classical and extended approaches, maturity terminology, and the operational definition adopted in this thesis. It then introduces eHealth and digital-health foundations before examining healthcare Digital Twin concepts and applications. Part III describes the structured literature-review methodology and develops the dual state of the art: Digital Twins in eHealth and healthcare, then cardiovascular Digital Twins. This part includes two comparative landscape tables, analytical synthesis, research gaps, and explicit implications for CardioTwin. Part IV proposes a conceptual architecture for Digital Twin-based eHealth systems. Part V concludes the thesis and presents the transition toward the CardioTwin PFE.

Part II

Background

CHAPTER 2

DIGITAL TWIN FUNDAMENTALS

2.1 Origin and Historical Evolution of the Digital Twin

The Digital Twin concept emerged from Product Lifecycle Management (PLM), where Michael Grieves introduced the idea of connecting a physical product with a corresponding virtual information construct across the product lifecycle [1–3]. In this early formulation, the central concern was not healthcare, Artificial Intelligence, or real-time analytics, but the possibility of maintaining a coherent relation between a manufactured product, the information describing it, and the processes that use that information.

The concept was later adopted in aerospace, manufacturing, cyber-physical systems, data science, and healthcare. NASA-oriented definitions emphasized high-fidelity simulation, sensor updates, maintenance history, and the need to mirror the operational life of a physical vehicle or system [4, 5]. Manufacturing and cyber-physical-system literature then strengthened the role of synchronization, data exchange, simulation, and production services [6–10]. More recent reviews and taxonomies show that the Digital Twin is no longer interpreted only as a virtual product model, but also as a data-driven and service-oriented system whose configuration depends on the application domain [11–15].

Tomczyk and van der Valk [15] describe this evolution as a paradigm shift in the definition of the Digital Twin. Their analysis shows a movement from the classical three-dimensional view, composed of physical space, virtual space, and a bidirectional connection, toward an extended interpretation in which data and services become explicit dimensions. Figure 2.1 summarizes this historical movement as an author synthesis. The figure is not a reproduction of a published diagram; it organizes the definition literature reviewed in this thesis.

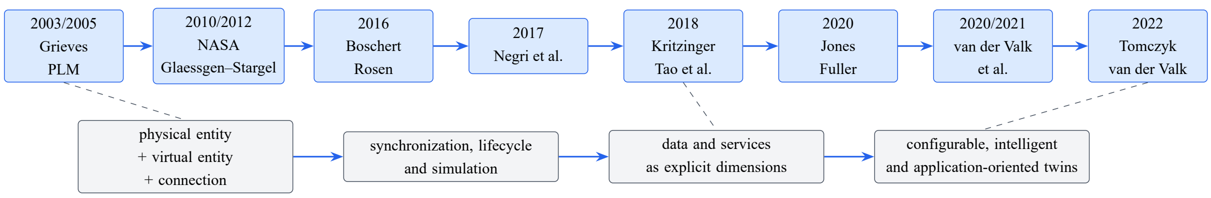

Figure 2.1: Evolution of representative Digital Twin definitions from lifecycle-oriented models to data- and service-centered systems. Author synthesis based on the reviewed literature.

The evolution shown in Figure 2.1 is important for eHealth because healthcare Digital Twins cannot be reduced to a three-dimensional visual model. They require traceable data, longitudinal updating, clinically meaningful services, governance, and validation. The following sections therefore examine the main definition streams before adopting the operational definition used in this thesis.

2.2 The Absence of a Universal Definition

There is no universally accepted Digital Twin definition. Tomczyk and van der Valk [15] argue that this plurality is partly explained by the interdisciplinary use of the concept and by the evolution of Information and Communication Technologies. A definition developed for PLM does not necessarily emphasize the same elements as a definition developed for aerospace, manufacturing, information systems, data science, smart cities, public health, or clinical medicine.

Definitions differ because the represented physical entity, application objective, required fidelity, data-exchange direction, synchronization frequency, expected services, available technologies, lifecycle stage, and degree of automation differ. For example, an aerospace Digital Twin may require detailed multiphysics simulation and maintenance history, while a hospital-process twin may focus on patient flow, resource allocation, and operational scenarios. A cardiovascular twin may combine ECG, imaging, hemodynamics, clinical history, and mechanistic models, while a public-health twin may represent aggregated populations and environmental factors.

Technological change also affects the definition. The rise of IoT, cloud computing, Big Data platforms, Artificial Intelligence, edge computing, service-oriented architectures, real-time analytics, and healthcare interoperability standards has expanded what a Digital Twin can do. As a result, recent literature increasingly treats data, services, connectivity, interfaces, and user-oriented functions as central elements rather than secondary implementation details [10, 12, 15]. This does not mean that one modern definition is universally correct. Instead, different definitions emphasize different Digital Twin functions, and the maturity of a system must be evaluated against its stated purpose.

2.3 Definitions of the Digital Twin in the Literature

This section presents representative author-attributed definitions. Each definition is discussed in relation to its original domain, central idea, data-exchange assumptions, service orientation, and limitation when transferred to healthcare.

2.3.1 Grieves' Product-Lifecycle Definition

Grieves conceptualizes the Digital Twin as a PLM construct linking a physical product, a virtual product, and the information connection between the two [1, 2]. The original domain is industrial product lifecycle management. The central idea is that information about a physical product should be available through a virtual counterpart across design, production, operation, and support.

The strength of this view is its clear physical–virtual structure. It establishes that a Digital Twin must refer to a specific physical counterpart and not merely to a generic model. However, the early PLM framing was strongly influenced by manufactured products and lifecycle information. When transferred to healthcare, it must be extended to account for biological variability, uncertainty, clinical interpretation, patient privacy, and human-supervised decision-making.

2.3.2 The NASA and Glaessgen–Stargel Definition

The NASA roadmap and the definition developed by Glaessgen and Stargel place the Digital Twin in the context of aerospace systems [4, 5]. Glaessgen and Stargel define the concept through an as-built vehicle or system supported by integrated multiphysics, multiscale, and probabilistic simulation, updated by sensor data, maintenance history, and fleet information. The objective is to mirror the operational life of the physical counterpart and support reliability, safety, and decision-making.

This definition is valuable because it emphasizes high-fidelity simulation, historical data, sensor updating, and uncertainty. Its limitation for healthcare is that patients are not engineered vehicles with fully specified designs, controlled operating conditions, or complete maintenance records. A patient twin may use simulation and sensor updates, but it must also handle incomplete observations, heterogeneous data, disease evolution, treatment response, and ethical constraints.

2.3.3 Boschert and Rosen's Lifecycle-Oriented Definition

Boschert and Rosen approach the Digital Twin from the simulation perspective [6]. Their lifecycle-oriented view emphasizes a comprehensive physical and functional description of the system and information useful during current and future lifecycle stages. Simulation is treated as an essential mechanism for exploring system behavior.

The relevance to healthcare lies in the longitudinal dimension. A patient or organ twin should not be a one-time model; it should preserve information over time and support future interpretation. However, lifecycle simulation alone is insufficient. Healthcare twins also require clinical validation, traceable provenance, safety-aware services, and careful distinction between measured, estimated, predicted, and simulated values.

2.3.4 Negri et al.'s Synchronization-Oriented Definition

Negri et al. [7] review the role of Digital Twins in cyber-physical production systems. Their interpretation emphasizes a virtual representation of a physical system, simulation across disciplines, synchronization with the real system, sensor data, connected smart devices, mathematical models, and real-time data processing.

This definition helps distinguish synchronized twins from static simulations. It makes the physical–virtual relationship operational by linking the virtual model to sensor-driven updates.

For healthcare, the synchronization principle is essential, but it must be adapted to different temporal scales: ECG may be sampled at high frequency, while demographics, medical history, and diagnosis labels change more slowly.

2.3.5 Kritzinger et al.'s Data-Exchange Definition

Kritzinger et al. [8] provide one of the most useful maturity distinctions in the Digital Twin literature by classifying digital representations according to the direction and automation of data exchange. A Digital Model involves manual data exchange between the physical and virtual spaces. A Digital Shadow includes automatic physical-to-digital data flow, but no equivalent automatic return flow. A Digital Twin requires automatic bidirectional exchange.

This classification is widely used because it prevents overuse of the term Digital Twin. Nevertheless, it does not by itself assess clinical validity, model fidelity, service usefulness, safety, or ethical acceptability. In healthcare, a bidirectional data link is not enough; the returned information must also be clinically meaningful, validated, explainable, and normally mediated by healthcare professionals.

2.3.6 Enders and Hoßbach's Information-Oriented Definition

Enders and Hoßbach [16] analyze Digital Twin applications across industries and define the concept as a virtual representation of a physical twin that may be connected to the physical counterpart and can provide additional information about it. Their information-oriented definition is useful because it highlights the informational value of the twin rather than treating the virtual entity as a purely geometric model.

The advantage of this view is flexibility across domains. Its weakness is that allowing the connection to be optional can blur the boundary between a Digital Twin and a disconnected digital model. For healthcare, the connection should be explicit and traceable because clinical interpretation depends on knowing where a value came from, how recently it was updated, and whether it is measured, inferred, predicted, or simulated.

2.3.7 Tao et al.'s Five-Dimensional Definition

Tao et al. [9, 10] extend the classical model by defining five dimensions: physical entity, virtual entity, Digital Twin data, services, and connections. This approach makes data and services independent dimensions rather than implicit by-products of the physical–virtual link. Qi and Tao [17] further relate Digital Twins to Big Data and Industry 4.0, while Qi et al. [18] emphasize Digital Twin services for smart manufacturing.

The five-dimensional view is important for eHealth because healthcare twins depend on multimodal data, longitudinal storage, interfaces, alerts, reporting, simulation, prediction, and decision support. Its limitation is that the presence of five architectural dimensions does not automatically guarantee maturity. A system may contain data and services but still lack validation, synchronization, or clinical credibility.

2.3.8 Jones et al.'s Characteristic-Based Definition

Jones et al. [11] characterize Digital Twins through recurring properties identified in the literature. These include the physical entity, virtual entity, physical environment, virtual environment, parameters, state, fidelity, physical-to-virtual connection, virtual-to-physical connection, twinning, physical process, and virtual process.

This characteristic-based approach is useful for comparing heterogeneous systems because it avoids reducing the Digital Twin to one rigid architecture. It also supports maturity analysis: two systems may both be called Digital Twins but differ strongly in fidelity, state representation, twinning rate, or feedback direction. In healthcare, such granularity helps distinguish an ECG classifier, an offline cardiac simulation, a digital shadow, and a connected patient-state twin.

2.3.9 van der Valk et al.'s Configurable Taxonomy

Van der Valk et al. [13] propose a multidimensional taxonomy of Digital Twins, and van der Valk et al. [14] later identify archetypes of Digital Twin configurations. Their work treats Digital Twins as configurable systems whose properties depend on use case, purpose, data link, model accuracy, synchronization, interface, data input, human–machine interaction, and machine-to-machine interaction.

The configuration-oriented view is especially relevant to healthcare. A hospital-process twin, a patient-monitoring twin, and a cardiovascular electrophysiology twin do not need identical fidelity, data frequency, or services. However, the taxonomy also implies a responsibility: each implementation should state its configuration clearly rather than using the Digital Twin label without specifying connection, purpose, synchronization, and service maturity.

2.3.10 Fuller et al.'s Enabling-Technology Perspective

Fuller et al. [12] review enabling technologies, challenges, and open research issues, including IoT, Artificial Intelligence, Machine Learning, simulation, cloud computing, and cybersecurity. Their perspective reinforces that a Digital Twin is not equivalent to any single technology. A dashboard, standalone simulation, disconnected 3D model, or isolated AI predictor may support a twin, but it is not itself a complete Digital Twin.

For healthcare, this distinction is essential. AI may support diagnosis, prognosis, personalization, and anomaly detection, but the Digital Twin maturity still depends on physical–virtual coupling, traceable data exchange, state maintenance, validation, and clinically meaningful services.

2.3.11 Tomczyk and van der Valk's Definition Synthesis

Tomczyk and van der Valk [15] synthesize Digital Twin definitions into three principal streams: the classical approach, the extended approach, and the application-oriented approach. Their concept-matrix analysis shows that newer definitions increasingly emphasize data, services, connectivity, Information and Communication Technologies, and user-oriented applications.

Their contribution is central to this thesis because it explains why definition plurality is not merely a terminological problem. It reflects the movement of Digital Twin research from product-centered representations toward connected systems that produce services through data, models, interfaces, and domain-specific configurations. In eHealth, this shift justifies analyzing not only whether a virtual patient model exists, but also whether it is updated, traceable, useful, validated, and safely integrated into clinical decision processes.

2.4 Comparative Analysis of Digital Twin Definitions

Table 2.1 compares the representative definitions discussed above. The criteria are selected to clarify why authors use different definitions and how each definition transfers, or fails to transfer, to healthcare. The table distinguishes original domain, definition focus, core elements, data-exchange assumptions, service purpose, and healthcare limitation.

Table 2.1: Comparison of representative Digital Twin definitions in the literature.

Author and year	Original domain	Definition focus	Core elements	Data exchange	Services or purpose	Main limitation for healthcare
Grieves [1, 2]	Product lifecycle management	Virtual product information linked to a physical product	Physical product, virtual product, connection, lifecycle information	Bidirectional link is conceptually central but initially product-information oriented	Lifecycle support, design, production, maintenance, decision support	Industrial product focus; requires extension for biological variability, privacy, clinical risk, and uncertainty
Shafto et al. and Glaessgen–Stargel [4, 5]	Aerospace and NASA vehicle systems	High-fidelity simulation of an as-built system	Multiphysics models, multiscale simulation, sensor updates, history, fleet data	Sensor and historical data update the virtual counterpart	Reliability, health management, mission support, life prediction	Assumes engineered systems and rich physical models; not directly sufficient for patient twins
Boschert and Rosen [6]	Mechatronics and simulation	Simulation across lifecycle stages	Physical and functional description, lifecycle information, simulation	Data support lifecycle simulation and future stages	Behavior exploration, design, operation, maintenance	Strong simulation focus but less explicit on clinical governance and human-supervised feedback
Negri et al. [7]	CPS-based production systems	Synchronization with real production systems	Virtual representation, sensors, smart devices, mathematical models, real-time processing	Sensor-driven synchronization with the physical system	Monitoring, simulation, control, production optimization	Useful for synchronization, but clinical time scales and validation requirements differ
Kritzinger et al. [8]	Manufacturing literature classification	Direction and automation of data exchange	Digital Model, Digital Shadow, Digital Twin	Manual, one-way automatic, or bidirectional automatic exchange	Terminological classification and maturity distinction	Does not alone capture clinical validity, fidelity, services, safety, or ethics
Enders and Hoßbach [16]	Cross-industry information systems	Information value of a virtual representation	Physical twin, digital representation, possible connection, additional information	Connection may exist but is not always mandatory	Additional information about the physical counterpart	Optional connection can blur the boundary between model and twin
Tao et al. [9, 10]	Smart manufacturing and Industry 4.0	Five-dimensional Digital Twin	Physical entity, virtual entity, data, services, connections	Multiple data and service connections across physical and virtual spaces	Evaluation, prediction, optimization, service delivery, lifecycle management	Strong architecture, but healthcare requires clinical validation and governance beyond architecture
Jones et al. [11]	Manufacturing definition review	Recurring Digital Twin characteristics	Entity, environment, parameters, state, fidelity, connections, twinning, processes	Physical-to-virtual and virtual-to-physical connections plus twinning rate	Comparative characterization of heterogeneous twins	Broad characterization must be translated into clinical evidence criteria

Continued on next page

Table 2.1: Comparison of representative Digital Twin definitions in the literature (continued).

Author and year	**Original domain**	**Definition focus**	**Core elements**	**Data exchange**	**Services or purpose**	**Main limitation for healthcare**
van der Valk et al. [13, 14]	Information systems and industrial taxonomy	Configurable Digital Twin taxonomy and archetypes	Purpose, data link, model accuracy, synchronization, interface, input, HMI, M2M interaction	Depends on configuration and use case	Use-case-specific configuration and archetype identification	Flexible but requires explicit reporting to prevent vague Digital Twin claims
Fuller et al. [12]	Cross-domain technology review	Enabling technologies and challenges	IoT, AI, ML, simulation, cloud, data analytics, cybersecurity	Technology-dependent physical–virtual integration	Monitoring, prediction, optimization, decision support	Technology focus may be misread as meaning that any AI or IoT system is a Digital Twin
Tomczyk and van der Valk [15]	Definition synthesis	Paradigm shift in Digital Twin definitions	Classical, extended, and application-oriented definition streams	Moves from bidirectional link toward broader connection, data, and services	Explains definitional evolution and concept-matrix patterns	Provides conceptual synthesis; implementation validity remains domain-specific

The comparison shows that Digital Twin definitions converge on a physical–virtual relationship but diverge on what must be explicit. Classical definitions identify the minimum structure. Simulation-oriented definitions strengthen model fidelity. Data-exchange definitions clarify the difference between models, shadows, and twins. Five-dimensional definitions make data and services visible. Taxonomies and archetypes recognize that Digital Twins are configured for use cases. For eHealth, the most appropriate interpretation must combine all these insights: a healthcare twin needs a physical counterpart, a virtual representation, traceable data, explicit connection, synchronization, services, validation, and human-supervised clinical use.

2.5 The Classical Three-Dimensional Approach

The classical approach describes a Digital Twin through three dimensions: physical entity, virtual entity, and bidirectional connection [2, 3, 15]. Figure 2.2 presents this structure as an author synthesis.

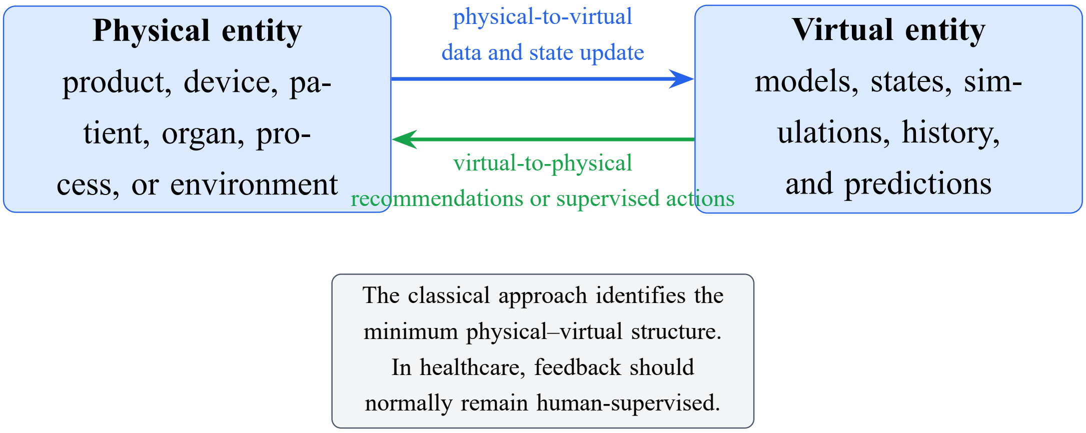

Figure 2.2: Classical three-dimensional Digital Twin approach. Author synthesis based on Grieves and related definitions.

Physical Entity

The physical entity is the real product, system, process, device, patient, organ, or environment represented by the twin. It has an observable state, behavior, operating environment, lifecycle, and potential interventions. In healthcare, the physical entity may be a patient, cardiovascular system, ECG acquisition device, care pathway, or hospital service.

Virtual Entity

The virtual entity is the digital counterpart. It may contain geometry, state variables, mathematical models, simulation, behavior, history, and prediction capabilities. It is not necessarily a three-dimensional visual model. A virtual entity can be a computational model, a state representation, a data structure, a simulation environment, or a combination of these elements.

Bidirectional Connection

The connection links the physical and virtual entities. The physical-to-virtual flow transfers measurements, events, and state updates. The virtual-to-physical flow returns information, recommendations, optimized parameters, control signals, or supervised actions. In healthcare, feedback from the virtual side should normally remain human-supervised because automated clinical control raises safety, ethical, and regulatory concerns. Under the Kritzinger taxonomy [8], a system with only automatic physical-to-virtual flow is better classified as a Digital Shadow than as a full Digital Twin.

2.6 The Extended Five-Dimensional Approach

The five-dimensional approach extends the classical model by adding Digital Twin data and services as explicit dimensions [9, 10, 15, 17]. Figure 2.3 summarizes the approach.

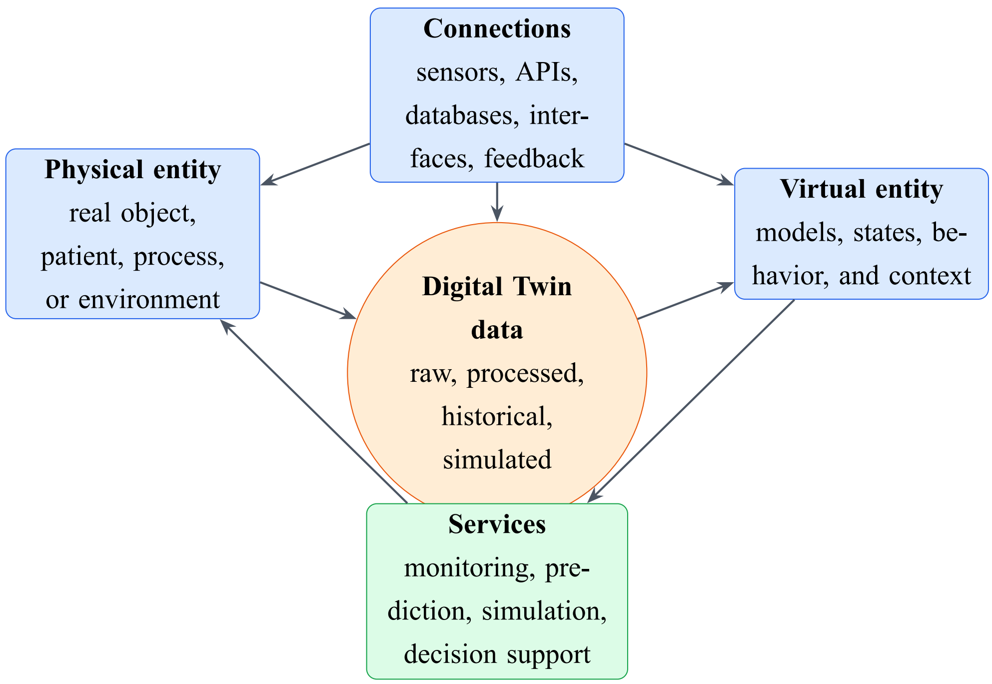

Figure 2.3: Extended five-dimensional Digital Twin approach. Author synthesis based on Tao et al. and Tomczyk and van der Valk.

Physical Entity

The physical entity remains the real-world object, patient, process, system, or environment. In eHealth, it includes not only the biological entity but also its clinical context, sensors, care setting, and interactions with healthcare professionals.

Virtual Entity

The virtual entity contains models, state descriptions, behavior representations, context, and simulations. It may include mechanistic models, statistical models, Machine Learning models, rule-based logic, or hybrid representations.

Digital Twin Data

Digital Twin data include raw measurements, processed signals, historical records, contextual variables, simulated values, model parameters, metadata, provenance, and uncertainty indicators. Data deserve a separate dimension because they are not merely transmitted between physical and virtual spaces; they are stored, transformed, validated, reused, and interpreted by services.

Connections

Connections include more than a single arrow. They cover sensors to acquisition modules, databases to virtual models, model services to interfaces, physical-to-virtual flows, virtual-to-physical flows, machine-to-machine communication, human-to-machine interfaces, APIs, and healthcare interoperability standards such as HL7 FHIR.

Services

Services transform the internal capabilities of a twin into useful functions. They include monitoring, visualization, diagnosis support, prediction, optimization, simulation, scenario analysis, validation, alerting, reporting, and decision support. In healthcare, services must be evaluated not only for technical performance but also for safety, usability, explainability, and clinical relevance.

2.7 From Classical to Extended Digital Twin Definitions

The extended five-dimensional approach does not reject the classical approach. Instead, it makes two previously implicit elements explicit: data and services. Table 2.2 compares both approaches and their implications for eHealth.

Table 2.2: Comparison between the classical three-dimensional and extended five-dimensional Digital Twin approaches.

Criterion	**Classical three-dimensional approach**	**Extended five-dimensional approach**	**Implication for eHealth**
Represented entity	Physical object or system	Physical entity plus environment and context	Patient, organ, device, workflow, or population context must be explicit
Virtual model	Digital counterpart of the physical entity	Virtual entity supported by data, models, and services	Clinical state representation may combine models, records, signals, and simulations
Connection	Bidirectional physical–virtual link	Multiple connections among sensors, databases, services, and interfaces	Requires interoperable and traceable healthcare data flows
Role of data	Often implicit in the connection	Independent dimension	Multimodal and longitudinal data management becomes central
Role of services	Often implicit as use of the model	Independent dimension	Dashboards, alerts, reports, simulation, and decision support must be evaluated
User interfaces	Not always emphasized	Part of service and connection logic	Clinicians and patients need understandable outputs
Scalability	Usually focused on one product or system	Platform and service orientation	Health systems need modular architectures and integration with existing systems
Lifecycle	Product lifecycle focus	Lifecycle plus evolving services and data	Patient state, disease trajectory, and treatment history must be maintained
Interoperability	Not central in early definitions	Connection and data dimensions make it explicit	Standards such as FHIR become architectural requirements
Decision support	Possible but not always specified	Service dimension makes it explicit	Clinical recommendations require validation and human supervision
Healthcare suitability	Useful minimum structure	Better suited to complex eHealth platforms	Supports multimodal data, AI, simulation, governance, and clinical services

The classical approach remains useful because it identifies the minimum physical–virtual structure. The extended approach is more appropriate for complex eHealth platforms because healthcare systems require multimodal data, longitudinal storage, interoperability, AI, simulation, dashboards, alerts, clinical services, and governance. However, neither approach is sufficient unless the maturity and validation of the implementation are also assessed.

2.8 Application-Oriented and Configuration-Oriented Approaches

Some Digital Twin definitions are derived from specific use cases rather than from a universal architecture. Examples include manufacturing systems, aerospace vehicles, hospital workflows, individual patients, organs, medical devices, public-health processes, and cardiovascular models. Application-oriented definitions have practical advantages: they provide domain relevance, clear services, measurable validation targets, and implementation guidance.

Their risk is definition fragmentation. If each domain defines Digital Twin only around its own needs, the term may be used for any simulation, dashboard, or prediction system. The taxonomy of van der Valk et al. [13, 14] addresses this problem by treating Digital Twins as configurable systems. A Digital Twin can be configured differently depending on purpose, data link, model accuracy, synchronization, interface, data input, and interaction mode, but the configuration should be reported transparently.

Figure 2.4 summarizes the three definition streams highlighted by Tomczyk and van der Valk [15]. The figure helps interpret healthcare Digital Twins: they are application-oriented, but they should still satisfy general physical–virtual, data, connection, and service criteria.

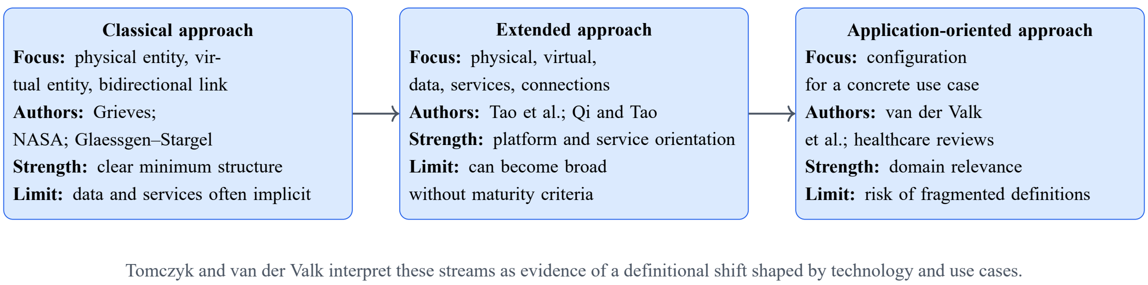

Figure 2.4: Three principal streams of Digital Twin definitions identified in the literature. Author synthesis based on Tomczyk and van der Valk.

2.9 Digital Model, Digital Shadow, and Digital Twin

The distinction between Digital Model, Digital Shadow, and Digital Twin is based on two dimensions: the automation of data exchange and the direction of data exchange [8]. This distinction is essential in healthcare because many systems described as Digital Twins are actually static models, episodic simulations, or one-way monitoring systems.

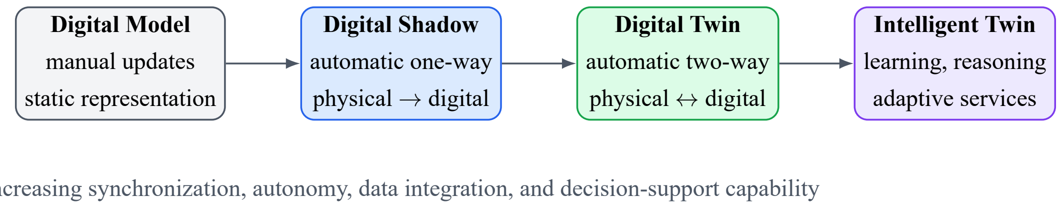

Figure 2.5: Evolution from digital model to intelligent Digital Twin. Author synthesis based on the reviewed literature.

Table 2.3 clarifies the distinction and adds the intelligent or cognitive Digital Twin direction. An intelligent Digital Twin may include learning, reasoning, adaptation, optimization, and uncertainty-aware services, but AI alone does not transform a model into a Digital Twin. Physical–virtual connection and state updating remain necessary.

Table 2.3: Distinction between Digital Model, Digital Shadow, Digital Twin, and intelligent Digital Twin.

System type	Physical-to-virtual exchange	Virtual-to-physical exchange	Automatic update	Typical service	Main limitation
Digital Model	Manual or absent	Manual or absent	No	Offline analysis or documentation	Static and disconnected from current physical state
Digital Shadow	Automatic from physical to virtual	Manual or absent	One-way update	Monitoring, dashboards, retrospective analysis	Lacks automatic bidirectional influence or controlled feedback
Digital Twin	Automatic and traceable	Automatic or supervised feedback	Bidirectional update according to purpose	Monitoring, simulation, prediction, decision support	Requires validation, governance, and safe service integration
Intelligent Digital Twin	Automatic and learning-aware	Adaptive recommendations or optimized actions	Bidirectional plus model adaptation	Personalization, optimization, uncertainty-aware support	AI may drift, overfit, or become opaque without revalidation

2.10 Fundamental Characteristics of a Digital Twin

The definition literature repeatedly identifies characteristics that can be mapped to the five-dimensional model. Jones et al. [11] emphasize entity, environment, state, parameters, fidelity, connections, twinning, and processes. Van der Valk et al. [13, 14] add configuration dimensions such as purpose, data link, synchronization, model accuracy, data input, and interfaces. Figure 2.6 organizes these characteristics into the five-dimensional view.

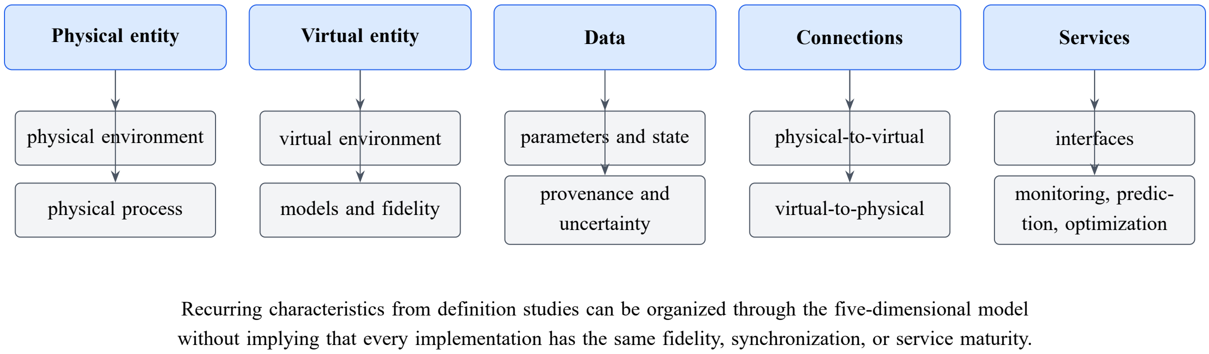

Figure 2.6: Mapping of recurring Digital Twin characteristics to the five-dimensional model. Author synthesis based on the reviewed definition studies.

This mapping shows why a Digital Twin should not be evaluated through a single criterion. A connected model may still have low fidelity. A high-fidelity simulation may lack synchronization. A dashboard may provide services without maintaining a virtual state. A mature healthcare twin must therefore be described through its entity, data, model, connection, synchronization, services, validation, and clinical context.

2.11 Digital Twin Fidelity, Synchronization, and Twinning Rate

Fidelity is the degree to which the virtual entity reproduces the aspects of the physical counterpart that matter for the intended purpose. It may include geometric fidelity, behavioral fidelity, physiological fidelity, temporal fidelity, and functional fidelity. Maximum fidelity is not always necessary or feasible. A hospital-flow twin does not need cardiac electrophysiology, while an ablation-planning twin may require detailed anatomy and conduction modeling.

Synchronization concerns how the virtual counterpart is updated in relation to the physical entity. Synchronization can be continuous, periodic, event-driven, episodic, or manual. The appropriate frequency depends on the use case. ECG acquisition may require high-frequency sampling; blood-pressure trends may be updated periodically; demographic data and medical history change more slowly.

The twinning rate describes how often and how quickly the physical and virtual counterparts are reconciled. It depends on latency, sampling rate, update frequency, computational cost, and clinical urgency [11, 13]. In healthcare, a high twinning rate is useful only when the data are reliable and the service requires rapid response. A slow but clinically validated update can be more useful than a fast but poorly calibrated model.

2.12 Digital Twin Data and Services

Digital Twin data can be measured, calculated, estimated, predicted, simulated, contextual, or metadata. Measured data come directly from sensors, devices, or clinical observations. Calculated data are derived through deterministic formulas. Estimated data infer hidden states or parameters. Predicted data describe expected future values. Simulated data result from what-if scenarios or model experiments. Metadata and provenance describe source, timestamp, unit, processing history, and uncertainty.

This distinction is especially important for eHealth and CardioTwin because clinical users should not interpret a simulated treatment scenario in the same way as a measured ECG or blood-pressure value. Appendix A therefore includes supplementary review dimensions and value-type traceability principles that support auditability and uncertainty communication.

Digital Twin services may be descriptive, diagnostic, predictive, prescriptive, simulation-based, monitoring-oriented, reporting-oriented, optimization-oriented, or decision-support services [10, 12]. Service maturity depends on data quality, model validity, synchronization, user interface, governance, and clinical evaluation. A technically advanced service that is not explainable, validated, or safely integrated into workflow may have limited clinical value.

2.13 Digital Twin Lifecycle

A Digital Twin is not only created; it evolves. Figure 2.7 presents the lifecycle from purpose definition to retirement as an author synthesis based on the reviewed literature.

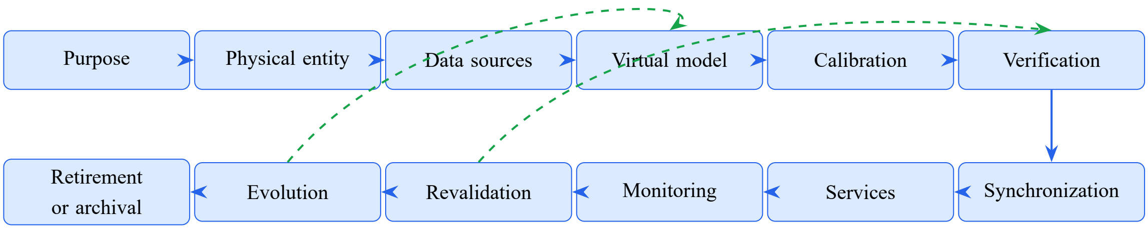

Figure 2.7: Lifecycle of a Digital Twin from purpose definition to operation, evolution, and retirement. Author synthesis based on the reviewed literature.

The lifecycle begins by defining the purpose of the twin and identifying the physical counterpart. It then requires data-source selection, virtual-model design, initialization, calibration, verification, synchronization, service delivery, monitoring, revalidation, evolution, and eventual retirement or archival. In healthcare, the twin evolves with disease progression, treatment, behavior, sensors, care setting, clinical objectives, and available data. These changes may require recalibration or revalidation, especially when the system supports clinical decisions.

2.14 Enabling Technologies

Digital Twin systems rely on several enabling technologies. IoT and IoMT devices support data acquisition from sensors, medical devices, and connected environments. Cloud computing provides scalable storage and processing, while edge computing can reduce latency and support local preprocessing. Big Data platforms manage large multimodal datasets. Artificial Intelligence and Machine Learning support pattern recognition, state estimation, prediction, personalization, anomaly detection, and surrogate modeling. Simulation provides interpretable and mechanistic representations of system behavior. APIs, middleware, and interoperability standards support communication among data sources, models, and services. Cybersecurity mechanisms protect data exchange, access control, auditability, and system integrity [10, 12, 19].

These technologies are enabling components, not definitions by themselves. A system does not become a Digital Twin merely because it uses AI, cloud computing, or IoT. The technologies must contribute to a maintained physical–virtual relationship and to services that are aligned with the purpose of the twin.

2.15 Digital Twin Maturity

Digital Twin maturity can be interpreted through connection, synchronization, service capability, intelligence, and validation. A low-maturity system may be a static digital model. A more advanced system may be a digital shadow that receives automatic data from the physical entity. A connected Digital Twin maintains bidirectional or human-supervised feedback. An intelligent Digital Twin adds learning, adaptation, optimization, or reasoning.

In healthcare, maturity must not be assessed only by technological complexity. A deep-learning model with high internal accuracy may still be only a prediction model if it is disconnected from a patient-state representation. Conversely, a clinically validated, synchronized, rule-based monitoring twin may be more mature for its purpose than an opaque AI system without external validation. The two states of the art in Chapter 6 therefore use cautious labels such as conceptual framework, static digital model, offline patient-specific simulation, episodically calibrated model, Digital Shadow, process Digital Twin, partial connected Digital Twin, and intelligent-DT direction.

2.16 Operational Definition Adopted in This Thesis

Based on the reviewed definitions, this thesis adopts the following operational definition:

> A Digital Twin is a purpose-oriented and continuously maintainable virtual representation of a physical healthcare entity or process, connected to its real counterpart through traceable data exchanges and capable of supporting validated monitoring, analysis, simulation, prediction, or decision-support services.

This definition requires a clearly identified physical counterpart, a virtual representation, traceable data, an explicit connection, updates or synchronization, purpose-oriented services, validation, and human supervision when clinical decisions are involved. It excludes standalone dashboards, disconnected simulations, one-time AI predictions, static digital records, and generic three-dimensional models without state updating.

For the rest of the thesis, this operational definition supports a cautious interpretation of healthcare and cardiovascular Digital Twins. CardioTwin is therefore treated as a practical cardiovascular specialization that should maintain a patient state, integrate measured and derived information, support ECG analysis and what-if scenarios, and clearly distinguish measured, calculated, estimated, predicted, and simulated values. It does not claim full clinical maturity without implementation, evaluation, and future clinical validation.

2.17 Conclusion

This chapter showed that the Digital Twin is a plural and evolving concept rather than a single universally fixed definition. Grieves' PLM view established the classical physical–virtual structure; NASA and aerospace definitions emphasized high-fidelity simulation; manufacturing literature clarified synchronization and data exchange; Tao et al. made data and services explicit; Jones et al. and van der Valk et al. provided characteristic and configuration perspectives; and Tomczyk and van der Valk synthesized the movement from classical definitions toward extended and application-oriented approaches.

The operational definition adopted in this thesis combines these insights and provides the terminology used in the healthcare and cardiovascular state-of-the-art analysis. The next chapter introduces the eHealth and digital-health environment in which these Digital Twin principles can be applied, before their healthcare-specific adaptation is examined in the following chapter.

CHAPTER 3
EHEALTH AND DIGITAL HEALTH FOUNDATIONS

3.1 Definition of eHealth

After the previous chapter established the conceptual foundations of Digital Twins, this chapter introduces the eHealth environment in which these principles can be applied. eHealth refers to the use of information and communication technologies to support healthcare services, medical decision-making, prevention, diagnosis, monitoring, treatment, administration, and patient engagement. It includes a wide range of systems such as telemedicine, mobile health, electronic health records, connected medical devices, remote monitoring platforms, and clinical decision support systems.

3.2 From Traditional Healthcare to Connected Healthcare

Traditional healthcare is often centered around episodic consultations, paper-based documentation, and hospital-centered workflows. Connected healthcare extends this model by enabling continuous or near-continuous data acquisition, remote communication, and digital coordination among patients, physicians, caregivers, and healthcare institutions.

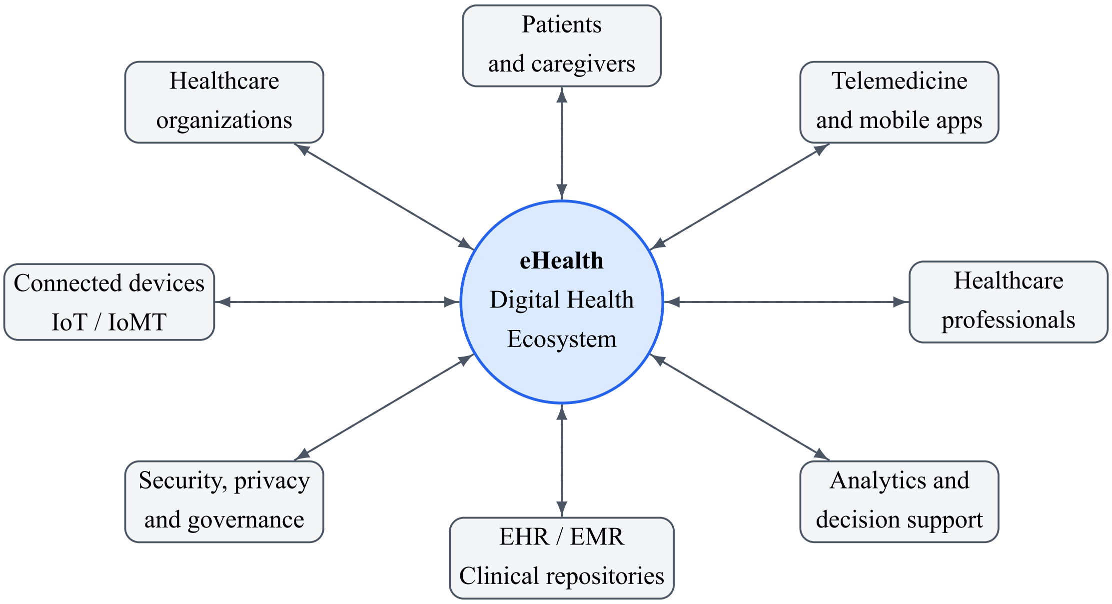

Figure 3.1: General eHealth ecosystem and its main stakeholders. Author synthesis.

3.3 Main Components of eHealth Systems

A general eHealth system may include several components:

- **Patient-facing technologies:** mobile applications, wearable sensors, home monitoring devices, and patient portals.
- **Clinical information systems:** EHR, EMR, laboratory information systems, radiology systems, and hospital management platforms.
- **Communication services:** teleconsultation, secure messaging, remote follow-up, and alerts.
- **Analytics and decision support:** machine learning models, rule-based reasoning, risk estimation, dashboards, and report generation.
- **Infrastructure:** cloud platforms, edge devices, APIs, databases, cybersecurity modules, and interoperability standards.

3.4 Data Sources in eHealth

eHealth systems rely on heterogeneous data sources. These include structured clinical data, medical measurements, biosignals, medical images, laboratory results, prescriptions, lifestyle data, environmental data, administrative data, and patient-reported outcomes. The combination of these sources is valuable but challenging due to differences in format, sampling frequency, quality, reliability, and privacy requirements.

3.5 Interoperability Standards

Interoperability is essential for eHealth because patient information may be generated and stored by different systems. Standards such as HL7 and FHIR support the exchange of healthcare information using common formats and resources. In a Digital Twin-based eHealth system, interoperability standards help connect data acquisition modules, clinical repositories, virtual models, and decision services.

3.6 Challenges in eHealth

The main challenges of eHealth include data fragmentation, missing data, inconsistent measurements, limited integration between systems, security and privacy risks, lack of clinical validation, usability barriers, and unequal access to digital services. These challenges justify the need for more integrated, dynamic, and trustworthy approaches such as Digital Twin-based architectures.

3.7 Conclusion

This chapter introduced the foundations of eHealth and highlighted the need for integrated and dynamic systems. Having introduced both Digital Twin fundamentals and the eHealth ecosystem, the next chapter explains how the Digital Twin paradigm is adapted to healthcare and eHealth applications.

CHAPTER 4

DIGITAL TWIN IN HEALTHCARE AND EHEALTH

4.1 Digital Twin for Health

A Digital Twin for Health inherits the general physical–virtual structure introduced in the Digital Twin fundamentals chapter, but it also depends on the eHealth ecosystem described in the previous chapter. Healthcare twins add domain-specific requirements that are particularly important in clinical contexts: they must preserve clinical meaning, integrate multimodal patient data, support longitudinal updating, communicate uncertainty, protect privacy, and remain subject to validation and human supervision. A Digital Twin for Health is therefore a virtual representation used to support healthcare-related monitoring, simulation, prediction, and decision-making while remaining connected to a clearly identified patient, organ, device, process, or health system.

Depending on its scope, it may represent a patient, an organ, a disease process, a medical device, a clinical workflow, a hospital unit, or a public health system. Unlike a simple dashboard, a Digital Twin aims to maintain a dynamic relationship between real-world data and virtual models. The general Digital Twin criteria still apply: a domain-specific health twin should not be considered mature merely because it contains medical data or an AI model.

4.2 Types of Healthcare Digital Twins

Healthcare Digital Twins can be classified into several categories:

- **Patient-specific Digital Twins:** virtual representations of individual patients using multimodal data.
- **Organ or body-system Digital Twins:** models focusing on a specific organ or physiological system.
- **Disease Digital Twins:** models representing disease progression, risk factors, or treatment response.

- **Medical device Digital Twins:** virtual representations of devices used for design, monitoring, or safety assessment.
- **Hospital and process Digital Twins:** models of workflows, resources, patient flow, logistics, and healthcare units.
- **Public health Digital Twins:** models used for population-level monitoring, planning, prevention, and crisis management.

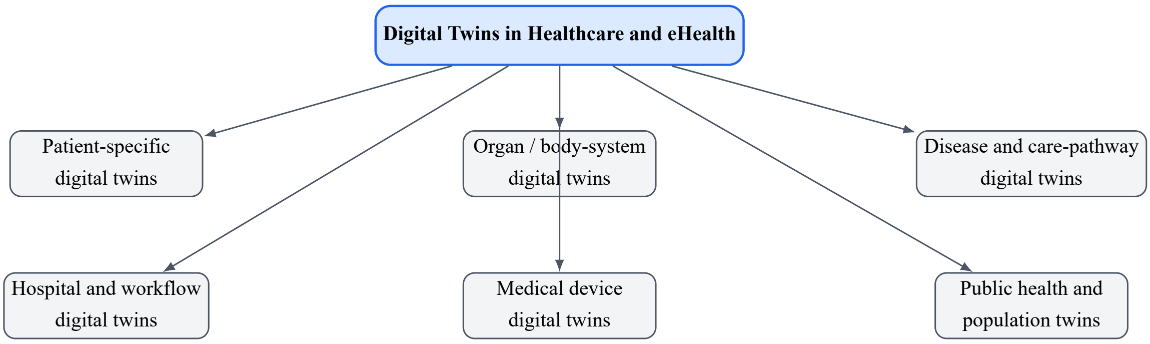

Figure 4.1: General taxonomy of Digital Twin applications in healthcare and eHealth. Author synthesis based on the reviewed literature.

4.3 Applications in eHealth

The applications of Digital Twins in eHealth are diverse. They include remote monitoring, personalized care pathways, treatment planning, rehabilitation support, chronic disease management, hospital resource optimization, medical device development, drug discovery, clinical trial simulation, and public health planning. These applications share a common objective: to transform heterogeneous data into actionable knowledge.

4.4 Patient-Centered eHealth Digital Twins

A patient-centered eHealth Digital Twin integrates clinical, physiological, behavioral, and contextual data to support personalized monitoring and decision-making. Such a system may receive information from EHRs, connected devices, laboratory results, patient-reported outcomes, and environmental sources. However, the level of clinical reliability depends on data quality, model validation, and the ability to handle uncertainty.

4.5 Healthcare Process Digital Twins

Digital Twins are not limited to individual patients. They can also represent healthcare processes, such as emergency department flow, outpatient services, operating room scheduling, hospital bed allocation, or vaccination campaigns. In this context, the Digital Twin supports simulation of operational scenarios and optimization of healthcare resources.

4.6 Ethical, Legal, and Social Considerations

Healthcare Digital Twins use sensitive data and may influence medical decisions. Therefore, their development must consider consent, privacy, data ownership, transparency, fairness, bias mitigation, accountability, and clinical responsibility. Ethical design is not an optional component; it is a necessary condition for trustworthy eHealth Digital Twins.

4.7 Challenges and Limitations

The implementation of Digital Twins in eHealth faces several challenges: interoperability between healthcare systems, access to high-quality multimodal data, real-time synchronization, computational cost, cybersecurity, explainability, regulatory approval, integration into clinical workflows, and acceptance by healthcare professionals and patients.

4.8 Conclusion

Digital Twins offer a promising direction for eHealth, but their adoption requires more than technical models. It requires architecture, governance, validation, security, interoperability, and ethical alignment. These issues are addressed in the state-of-the-art chapters that follow.

Part III

State of the Art

CHAPTER 5

LITERATURE REVIEW METHODOLOGY

5.1 Purpose and Review Design

This thesis uses a structured and focused state-of-the-art review. The objective is not to claim exhaustive coverage of every publication using the term Digital Twin, but to build a sufficiently broad and critically evaluated evidence base for two related domains: Digital Twins in eHealth and healthcare, and Digital Twins in cardiovascular medicine. The first domain establishes general concepts, architectures, services, enabling technologies, and governance requirements. The second provides a clinical deep dive into patient-specific modeling, electrophysiology, hemodynamics, heart failure, treatment simulation, and AI-assisted cardiovascular decision support.

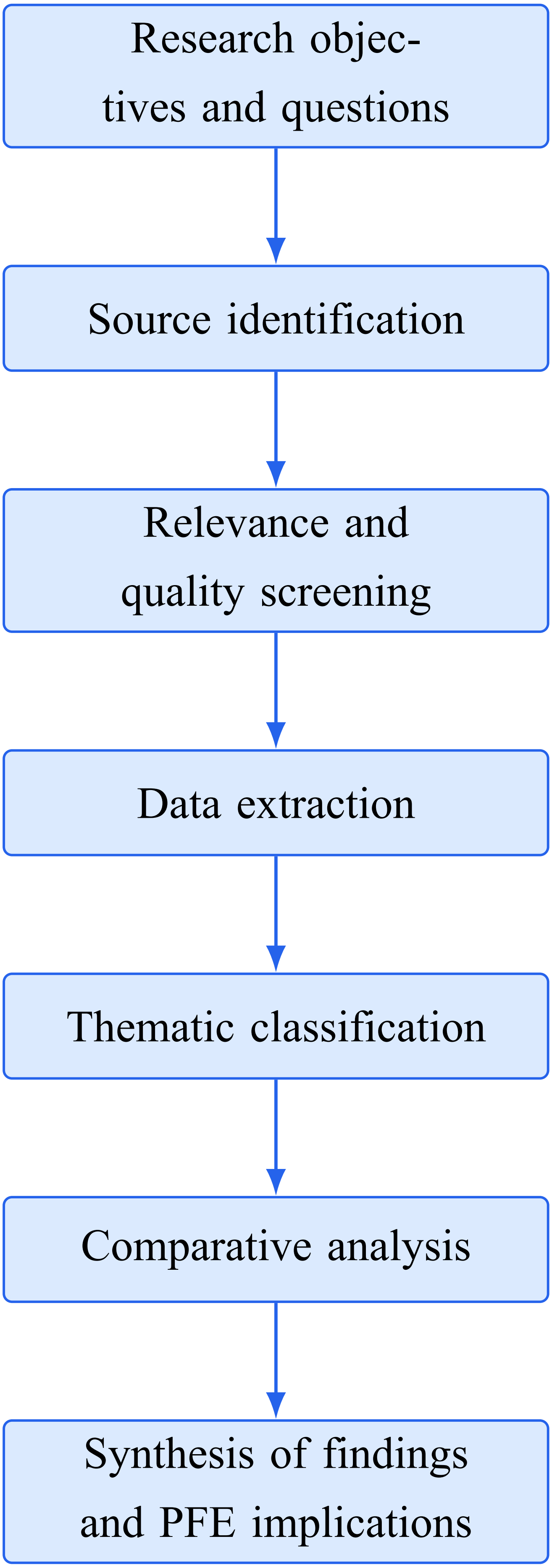

Figure 5.1: Workflow of the structured state-of-the-art literature review. Author synthesis.

5.2 Research Questions

- **RQ1:** How are Digital Twins defined and distinguished from digital models and digital shadows in healthcare?
- **RQ2:** What patient, organ, device, process, hospital, and public-health applications are reported in eHealth?
- **RQ3:** What architectures, data sources, AI methods, and simulation approaches are used in healthcare Digital Twins?
- **RQ4:** What forms of evidence and validation are reported for healthcare Digital Twins?
- **RQ5:** How are Digital Twins applied to cardiovascular electrophysiology, hemodynamics, heart failure, risk stratification, and treatment planning?
- **RQ6:** What research gaps prevent eHealth and cardiovascular twins from becoming continuously synchronized and clinically deployable systems?

5.3 Source Identification Strategy

Sources were identified through focused keyword searches, publisher and DOI pages, and backward citation tracing from major review papers. Search concepts included "Digital Twin definition", "Digital Twin three dimensions", "Digital Twin five dimensions", "Digital Model Digital Shadow Digital Twin", "Digital Twin for Health", "healthcare Digital Twin architecture", "human Digital Twin", "AI-powered Digital Twin healthcare", "cardiovascular Digital Twin", "cardiac electrophysiology Digital Twin", "heart failure Digital Twin", "virtual heart ablation", and "patient-specific hemodynamic model". General Digital Twin definition papers and AI methodology papers were included when their concepts were necessary for healthcare interpretation.

5.4 Inclusion and Exclusion Criteria

Studies were included when they provided at least one of the following: a healthcare-DT definition or taxonomy; an architecture; an original patient, organ, process, or public-health application; a cardiovascular patient-specific model with clinical data; a description of AI or hybrid modeling inside a twin; explicit validation evidence; or a substantive analysis of interoperability, uncertainty, ethics, regulation, or governance. Studies were excluded when the term Digital Twin was used only as promotional language, when the physical counterpart and model relationship were unclear, when the work lacked enough methodological detail for comparison, or when it duplicated a more complete source.

5.5 Data Extraction and Comparative Dimensions

For each source, the review extracted the represented physical entity, clinical objective, data modalities, modeling or architecture approach, synchronization direction, maturity, AI role, services, validation evidence, and stated limitations. Foundational definition sources were additionally analyzed by source definition, original domain, number and nature of Digital Twin dimensions, direction of data exchange, update frequency, fidelity, synchronization, service orientation, lifecycle orientation, and application dependence. These definition papers were included because inconsistent terminology directly affects the maturity assessment of healthcare and cardiovascular Digital Twins. AI-related sources were further classified by learning paradigm, model inputs, prediction or inference target, personalization level, explainability, uncertainty handling, and role inside the Digital Twin lifecycle. Cardiovascular studies were additionally analyzed by physiological scale, personalization data, calibration method, mechanistic assumptions, clinical comparator, cohort size when reported, and whether the system was a persistent twin or an offline patient-specific model.

5.6 Synthesis Strategy

The selected studies were synthesized in two states of the art. Within each domain, papers are first discussed thematically and critically, then compared in a dedicated landscape longtable. The synthesis does not rank papers using a single score because the studies pursue different goals and report different evidence. Instead, it identifies convergent architecture components, modeling patterns, maturity limitations, validation gaps, and requirements for future clinical translation. The synthesis also translates the literature findings into design implications for the CardioTwin PFE without treating the PFE itself as external scientific evidence.

5.7 Methodological Limitations

This is a structured state-of-the-art review rather than a PRISMA-compliant systematic review. It does not provide database-specific record counts, duplicate-removal statistics, or a meta-analysis. Publication terminology is heterogeneous, and the Digital Twin label is not used consistently. The review therefore applies cautious maturity labels and distinguishes conceptual frameworks, offline simulations, calibrated patient models, digital shadows, process twins, and connected Digital Twins.

5.8 Conclusion

This methodology supports a domain-level analysis of eHealth Digital Twins and a focused clinical analysis of cardiovascular Digital Twins. The following chapter applies the comparison dimensions to both corpora and develops separate comparative tables and research syntheses.

CHAPTER 6

DUAL STATE-OF-THE-ART REVIEW: DIGITAL TWINS IN EHEALTH AND CARDIOVASCULAR MEDICINE

6.1 Chapter Scope and Analytical Approach

This chapter develops two complementary states of the art. The first examines Digital Twins in eHealth and healthcare at the level of patients, organs, medical devices, care processes, hospitals, public health, and precision medicine. The second narrows the analysis to cardiovascular Digital Twins, where patient-specific computational models have been applied to electrophysiology, hemodynamics, heart failure, treatment planning, risk stratification, and virtual testing. The separation is deliberate: the eHealth review identifies the general concepts, architectures, data requirements, services, and governance conditions of healthcare Digital Twins, whereas the cardiovascular review evaluates how these principles are instantiated in one clinically demanding and data-rich domain.

The analysis combines review papers, architecture proposals, methodological studies, and original application studies. Each paper is examined according to the represented physical entity, input data, modeling strategy, physical–virtual synchronization, services, validation evidence, and limitations. The definition framework established in Chapter 3 is used throughout: a patient-specific simulation is not automatically a complete Digital Twin, and strong prediction performance does not replace continuous synchronization, model updating, traceability, or clinical validation. The chapter therefore distinguishes conceptual visions, static digital models, offline patient-specific simulations, episodically calibrated models, Digital Shadows, process Digital Twins, partial connected Digital Twins, and intelligent-DT directions.

6.2 State of the Art I: Digital Twins in eHealth and Healthcare

This first state of the art examines healthcare Digital Twins beyond a single disease or organ. It considers patient-specific twins, organ and body-system twins, disease and care-pathway twins, medical-device twins, hospital-process twins, and population-level twins. Figure 6.1 summarizes this application taxonomy as an author synthesis of the reviewed literature.

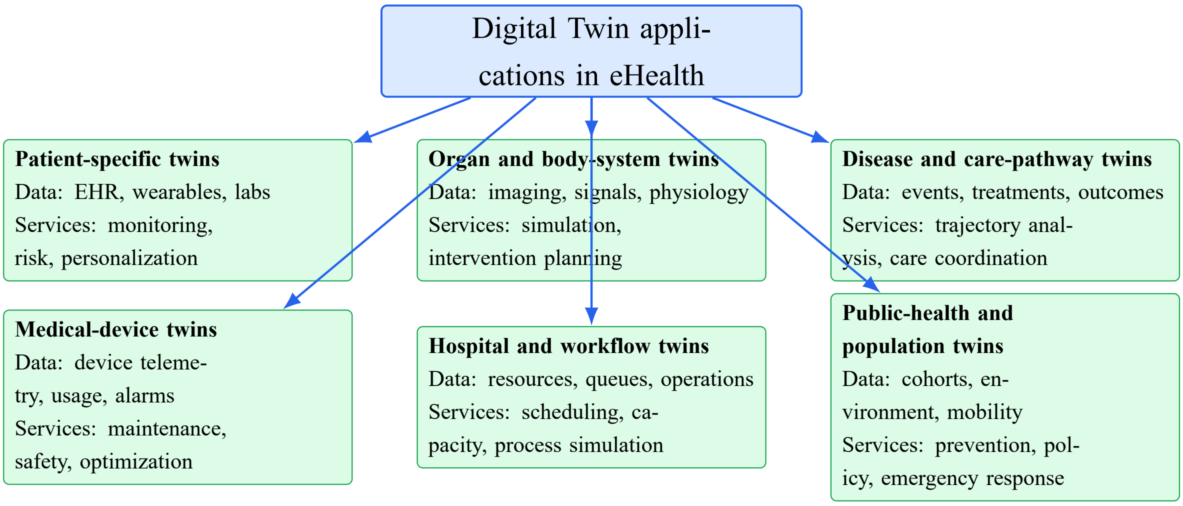

Figure 6.1: Taxonomy of Digital Twin applications in eHealth derived from the reviewed literature. Author synthesis based on the reviewed literature.

6.2.1 Evolution of the Healthcare Digital Twin Concept

The transfer of Digital Twin principles from engineering to healthcare introduced a fundamental change in the nature of the physical counterpart. Industrial twins usually represent assets with relatively well-defined boundaries and operating laws, whereas healthcare twins may represent a changing person, an organ, a disease trajectory, a medical device, a care pathway, or an organizational process. Human physiology is multiscale, partially observable, variable across individuals, and influenced by behavior, treatment, environment, and social context. Consequently, a healthcare Digital Twin cannot be assessed only by geometric fidelity or sensor connectivity; its credibility also depends on clinical meaning, uncertainty, longitudinal consistency, and safe integration into care workflows.

Katsoulakis et al. [20] provide one of the broadest mappings of Digital Twins for Health. Their scoping review organizes applications across hospital management, personalized medicine, surgical planning, medical-device development, drug discovery, in-silico trials, wellness, and public health. The authors also propose the five properties of an impactful health twin: individualized, interconnected, interactive, informative, and impactful. This formulation is valuable because it separates a genuine patient-centered twin from a generic computational model. However, the review also demonstrates that the field remains heterogeneous and that many systems described as Digital Twins lack dynamic bidirectional exchange or real clinical deployment.

Venkatesh et al. [21] frame health Digital Twins as precision-medicine tools created from multimodal patient data, population data, and continuously updated contextual variables. Their contribution is less architectural than translational: they analyze the computational, implementation, and regulatory requirements needed to move from a virtual patient concept toward clinical use. They emphasize that perturbing a digital representation to evaluate possible responses is attractive for precision medicine, clinical trials, and public health, but that validity, fairness, governance, and regulation remain central barriers.

Björnsson et al. [22] present a high-resolution personalized-medicine vision in which an individual model integrates molecular, cellular, clinical, and lifestyle information and is computationally exposed to many candidate treatments. This work is influential because it articulates treatment selection as a patient-specific modeling problem rather than a cohort-average decision. Nevertheless, it is primarily a conceptual and translational framework; the acquisition, synchronization, computational cost, and validation of such high-dimensional patient twins remain unresolved.

Tudor et al. [23] revisit human Digital Twin usage with a definition-based scoping review. Their analysis is especially useful for maturity assessment because it evaluates whether systems called human Digital Twins actually satisfy recognized Digital Twin characteristics. The study indicates that terminology remains inconsistent, application goals vary considerably, and only a subset of reported systems includes the persistent connection and updating expected from a mature twin.

6.2.2 Patient-Specific, Organ-Level, and Process-Oriented Twins

Healthcare Digital Twins can be separated into three broad implementation families. Patient-specific or human twins integrate heterogeneous information to represent an individual state and support personalized monitoring or intervention. Organ-level twins focus on a constrained physiological system and often rely on mechanistic models, imaging, and biosignals. Process-oriented twins represent hospital units, resource flows, public-health processes, or care pathways and are generally based on event data and discrete-event simulation rather than physiology.

De Benedictis et al. [24] classify healthcare applications and propose a generalized six-layer architecture that separates the physical twin, data, Digital Twin, services, connectivity, and security. Their CanTwin case demonstrates a process-oriented Digital Twin for social-distancing monitoring, queue analysis, people counting, and occupancy supervision. The study is important because it moves beyond a purely conceptual diagram and exposes architectural components in a real organizational setting. Its limitation is equally informative: success in a process twin does not directly establish the validity of a physiological patient twin, because the models, validation criteria, safety consequences, and uncertainty are different.

Noeikham et al. [25] derive a five-layer architecture for a healthcare unit using a structured literature review, software-design patterns, and a chemotherapy-center case study. Their architecture emphasizes modularity, scalability, security, privacy, and integration with Electronic Health Records and Clinical Data Repositories. The work shows that a healthcare-unit twin must connect to existing information systems and that process simulation can evaluate operational changes without disrupting real care. However, the case remains centered on service flow and cannot be generalized directly to patient physiology without additional modeling and clinical-validation layers.

Kamel Boulos and Zhang [26] broaden the scope from personalized medicine to precision public health. Their perspective highlights that Digital Twins can represent populations, places, and health systems as well as individuals. This extension is relevant to eHealth because disease prevention, resource allocation, and emergency response often require population-level simulation. At the same time, the aggregation of individual and environmental data introduces additional privacy, representativeness, and governance concerns.

6.2.3 Artificial Intelligence in Healthcare Digital Twins

Artificial Intelligence is an enabling component of a Digital Twin rather than its defining property. A prediction model may classify disease or forecast risk without maintaining a virtual counterpart of a specific physical entity. Conversely, a synchronized mechanistic or rule-based twin may operate without sophisticated learning. AI becomes especially valuable when the twin must process high-dimensional data, estimate hidden states, personalize parameters, approximate expensive simulations, recognize patterns, or forecast future trajectories.

Janiesch et al. [27] distinguish the broad field of AI from Machine Learning and Deep Learning. Their methodological account is useful for healthcare Digital Twins because it links model choice to data type, sample size, task complexity, computational resources, and interpretability. Conventional ML methods can remain appropriate for structured clinical data and smaller datasets, while deep neural networks are often better suited to images, biosignals, and complex temporal data. The most complex model is not automatically the best clinical model; out-of-sample performance, calibration, explanation, and operational constraints are equally important.

Chaparro-Cárdenas et al. [19] review the convergence of Digital Twins and AI for personalized and predictive healthcare. The paper covers AI-supported diagnosis, rehabilitation, multimodal fusion, predictive modeling, and adaptive services, while also stressing privacy, data heterogeneity, edge computing, federated learning, scalability, and clinical validation. A key contribution is the identification of the gap between technically impressive prototypes and systems capable of reliable real-time integration with clinical environments. The review supports the use of hybrid and multimodal intelligence but does not provide evidence that these approaches are already routine clinical practice.

Elgammal et al. [28] analyze 17 recent AI-powered Digital Twin applications across heart health, diabetes, mental health, respiratory care, stress management, and other domains. Their review shows that current systems employ conventional ML, deep learning, generative AI, Large Language Models, and graph-based methods for prediction, personalization, patient-trajectory analysis, and user interaction. It also reveals a maturity problem: many applications are retrospective, experimental, or based on synthetic data, and their clinical workflow integration, uncertainty, external validation, and regulatory status remain limited.

Sharma and Kaur [29] propose a hybrid patient-specific framework combining mechanistic physiological simulation, neural encoders, multimodal clinical data, Bayesian reasoning, and reinforcement learning. The framework directly addresses a central limitation of pure AI systems by grounding prediction in physiological structure. It also addresses limitations of purely mechanistic models by using learning for feature extraction and personalization. Nevertheless, the breadth of the proposed architecture exceeds the clinical evidence currently available; independent validation, prospective testing, and transparent reporting of data and model provenance are required before the approach can be considered clinically actionable.

Across these studies, AI contributes at several stages of the twin lifecycle: artifact detection and preprocessing, latent-state estimation, image segmentation, anomaly detection, diagnosis support, prognosis, patient stratification, parameter calibration, surrogate modeling, scenario ranking, and continuous adaptation. Each role creates a different validation obligation. An imputation model must not silently create clinically important values; a diagnostic model requires discriminative and calibration testing; an adaptive model requires drift monitoring and controlled revalidation; and a surrogate simulation requires evidence that it preserves the behavior of the mechanistic model inside the intended domain.

Figure 6.2 positions these roles across the healthcare Digital Twin lifecycle. The figure deliberately separates AI capability from Digital Twin maturity: the presence of learning algorithms can enrich a twin, but maturity still depends on the persistence of the physical–virtual relationship, controlled state updates, service integration, and validation.

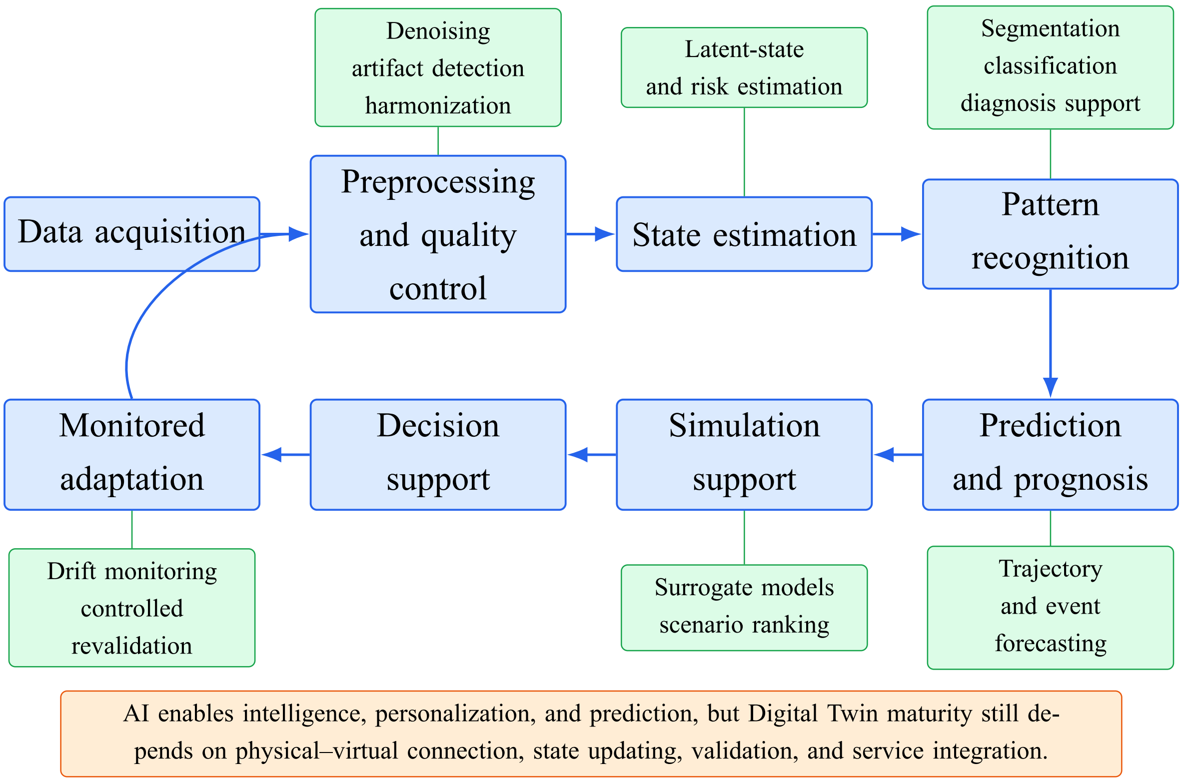

Figure 6.2: Functional roles of Artificial Intelligence across the healthcare Digital Twin lifecycle. Author synthesis based on the reviewed literature.

6.2.4 Interoperability, Validation, and Governance

Interoperability is a structural requirement rather than a peripheral feature. Patient twins may receive data from sensors, mobile applications, EHRs, imaging platforms, laboratories, and clinician inputs, each with different identifiers, formats, semantics, time scales, and data-quality characteristics. Healthcare standards such as HL7 FHIR can facilitate exchange, but syntactic compatibility alone does not guarantee semantic consistency or temporal alignment. The twin must preserve provenance, measurement units, timestamps, uncertainty, and the distinction between measured, calculated, estimated, predicted, and simulated values.

Meijer et al. [30] focus on methodological challenges and show that healthcare Digital Twins require explicit attention to causal assumptions, model updating, uncertainty, missing data, and validation. Their work is particularly valuable because it challenges the tendency to treat a high-performing predictor as a complete patient twin. They argue for methodological clarity regarding the target represented, the intervention modeled, and the evidence required to support individual-level claims.

Bruynseels et al. [31] analyze the ethical consequences of representing individuals through detailed, longitudinal virtual counterparts. They show that Digital Twins may alter the interpretation of health, normality, prevention, and enhancement by comparing a patient both with population patterns and with their own historical baseline. This creates opportunities for personalized prevention but also risks discrimination, unequal access, excessive surveillance, and opaque secondary use of data. Governance, consent, and transparency must therefore be embedded throughout the architecture.

Clinical credibility requires multiple forms of evaluation. Technical verification asks whether software, data pipelines, and numerical implementations behave as specified. Model validation examines whether outputs reproduce relevant physiological or operational behavior. External validation tests transportability to other populations, institutions, devices, and time periods. Prospective clinical validation evaluates whether the twin improves decisions or outcomes in real workflows. Human-factors evaluation assesses usability, alert burden, trust, and the effect of the system on clinical responsibilities. Most reviewed studies address only a subset of these layers.

6.2.5 Comparative Table for Digital Twins in eHealth

Table 6.1 compares representative eHealth and healthcare Digital Twin studies. The selected studies are not treated as an exhaustive inventory of all healthcare Digital Twin publications; rather, they were chosen because together they cover conceptual definitions, patient-specific visions, process-oriented twins, healthcare-unit architectures, public-health perspectives, AI-enabled approaches, methodological requirements, and ethics. The comparison criteria therefore emphasize the represented entity, data integration, modeling approach, maturity, evidence, and limitations. This structure makes it possible to compare conceptual reviews with implementations, patient twins with process twins, AI-only approaches with hybrid approaches, and architecture proposals with clinically oriented systems.

Table 6.1: Comparative state of the art of Digital Twins in eHealth and healthcare.

Study	Scope / represented entity	Data and integration	Modeling / architecture	Maturity	Contribution and evidence	Main limitations
Katsoulakis et al. [20]	Field-wide DT for Health review: patients, organs, hospitals, devices, trials, wellness, and public health	Clinical, molecular, imaging, sensor, environmental, and social data	Scoping review and 5I characterization of healthcare twins	Field-level synthesis	Broad taxonomy of applications and opportunities; distinguishes computational models from health twins	Heterogeneous terminology; many applications remain conceptual; limited standardization and clinical validation
Venkatesh et al. [21]	Precision medicine, clinical trials, and public health	Multimodal patient, population, environmental, and real-time data	Computational and regulatory perspective	Health-DT vision	Connects patient perturbation and simulation with implementation and regulation	No deployed system; uncertainty, fairness, governance, and regulation remain open
Björnsson et al. [22]	High-resolution individual patient model for treatment selection	Molecular, cellular, clinical, and lifestyle information	Systems medicine and large-scale virtual treatment testing	Conceptual patient twin	Influential personalized-medicine vision linking integrated patient models to drug selection	Very high data and modeling requirements; limited evidence for continuous synchronization and clinical workflow use
Tudor et al. [23]	Human Digital Twins across healthcare applications	Varies by application; clinical, sensor, imaging, and behavioral data	Definition-based scoping review	Maturity audit	Evaluates whether reported human twins comply with recognized DT characteristics	Reveals inconsistent terminology and limited numbers of fully connected, persistent twins
De Benedictis et al. [24]	Healthcare processes and public-health-oriented workplace monitoring	Organizational, location, occupancy, and process data	Generalized six-layer architecture and CanTwin case	Process Digital Twin	Provides explicit functional layers and a real case for social-distancing monitoring	Process-oriented; limited transfer to patient physiology, clinical outcomes, and patient-specific validation
Noeikham et al. [25]	Healthcare-unit service system and chemotherapy center	EHR, CDR, event, resource, and workflow data	Five-layer healthcare-unit architecture and discrete-event simulation	Process Digital Twin	Derives modular architecture with privacy, security, interoperability, and case analysis	Single unit-level case; physiological twinning and broader deployment remain untested
Kamel Boulos and Zhang [26]	Personalized medicine and precision public health	Individual, population, geospatial, and contextual data	Conceptual synthesis of patient and population twins	Conceptual framework	Extends DT thinking from individuals to population planning and public health	Limited implementation evidence; aggregation introduces privacy, bias, and representativeness issues
Meijer et al. [30]	Methodological foundations of healthcare twins	Longitudinal clinical and patient-specific data	Statistical, causal, and validation perspective	Methodological foundation	Clarifies causal assumptions, updating, uncertainty, and individual-level validity	Does not implement a specific twin; recommendations require operational benchmarks

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Table 6.1: Comparative state of the art of Digital Twins in eHealth and healthcare (continued).

Study	**Scope / represented entity**	**Data and integration**	**Modeling / architecture**	**Maturity**	**Contribution and evidence**	**Main limitations**
Chaparro-Cárdenas et al. [19]	AI-enhanced DTs for personalized and predictive healthcare	Biosignals, imaging, EHR, sensors, biomechanical and multimodal streams	Technological review of AI, hybrid reasoning, edge, cloud, and federated approaches	AI-enabled DT synthesis	Maps AI roles in prediction, diagnosis, rehabilitation, personalization, and adaptive services	Data heterogeneity, privacy, scalability, real-time integration, regulation, and controlled clinical testing
Elgammal et al. [28]	AI-powered applications across multiple health domains	EHR, wearables, glucose, imaging, behavioral, respiratory, and population data	ML, DL, generative AI, LLM, and graph-based methods	Review of practical and emerging applications	Identifies 17 applications across heart health, diabetes, mental health, respiratory care, and stress	Many studies remain retrospective or experimental; black-box behavior, interoperability, validation, and adoption barriers
Sharma and Kaur [29]	Patient-specific personalized healthcare	EHR, ECG, imaging, wearable, and clinical data	Hybrid mechanistic simulation, neural encoders, Bayesian inference, and reinforcement learning	Intelligent-DT framework	Proposes continuous synchronization, multimodal fusion, trajectory prediction, and therapy simulation	Broad claims require independent reproduction, external validation, prospective testing, and regulatory assessment
Bruynseels et al. [31]	Ethical implications of human and health twins	Longitudinal biological, physiological, and lifestyle data	Ethical and conceptual analysis	Governance foundation	Identifies implications for normality, prevention, enhancement, privacy, inequality, and discrimination	No technical implementation; ethical principles must be translated into enforceable architecture and policy

6.2.6 Synthesis of the eHealth State of the Art

The eHealth literature shows a progression from broad personalized-medicine visions toward more explicit architectures and application-specific prototypes. Yet the progression is uneven. Process twins currently offer some of the clearest examples of real connectivity because event streams and service metrics can update operational simulations. Patient twins are scientifically more ambitious: they must reconcile incomplete multimodal measurements with models of a biological system that changes over time and cannot be fully observed.

Five conclusions emerge. First, a healthcare twin must have a clearly defined physical counterpart and clinical purpose. Second, longitudinal data and model updating are essential for claims of personalization. Third, AI improves data interpretation and adaptation but does not establish Digital Twin maturity by itself. Fourth, interoperability, provenance, uncertainty, and governance must cross all architecture layers. Fifth, the evidence base remains dominated by reviews, conceptual frameworks, retrospective analyses, and focused prototypes; prospective demonstrations of clinical benefit are comparatively rare.

The table also shows that healthcare Digital Twin research is divided between broad conceptual frameworks and narrower technical systems. Patient-centered works emphasize personalization and clinical reasoning but often lack persistent synchronization, while process-oriented twins more clearly demonstrate connected data flows but represent organizational rather than physiological entities. AI-enhanced approaches expand diagnosis support, prediction, and personalization, yet their evidence is often retrospective or experimental. Hybrid AI–mechanistic proposals are scientifically attractive because they combine pattern recognition with physiological structure, but they create additional validation obligations. These observations are directly relevant to the PFE because CardioTwin must not be presented as an ECG classifier alone; it must be framed as a patient-state system with traceable data, controlled updates, interpretable services, and human-supervised outputs.

6.2.7 Implications of the eHealth State of the Art for CardioTwin

The general eHealth literature provides the architectural and methodological rationale for CardioTwin. It shows that a Digital Twin for healthcare cannot be reduced to a single AI module, dashboard, or prediction model. A clinically meaningful prototype must maintain a virtual patient state, update this state through validated data flows, preserve provenance, and distinguish between values that are measured, calculated, estimated, predicted, or simulated. This distinction is essential for CardioTwin because ECG analysis, blood-pressure measurements, risk scores, and what-if scenarios do not have the same evidential status.

The reviewed studies also justify the main CardioTwin components. Biomedical acquisition responds to the need for a physical–virtual connection. ECG processing and AI-assisted interpretation respond to the role of AI in biosignal analysis. SBP, DBP, heart-rate, clinical-history, and risk-factor integration support multimodal patient-state representation. Longitudinal storage and state updates address the requirement for persistent personalization. What-if scenarios and dashboards translate the service layer into practical decision support, while human-in-the-loop supervision, explainability, privacy, and validation remain cross-cutting constraints rather than optional additions.

Table 6.2: Design implications of the eHealth state of the art for CardioTwin.

Literature finding	**Implication for CardioTwin**	**Implementation direction**	**Limitation or precaution**
AI alone is not a Digital Twin	Maintain a persistent virtual patient state	Fuse ECG, BP, clinical variables, and historical information	This does not by itself prove full clinical Digital Twin maturity
Digital Twins require synchronization	Update the patient state after validated measurements	Sensor- or API-based acquisition and update pipeline	State quality depends on measurement reliability and timing
Multimodality supports personalization	Combine biosignals, SBP, DBP, symptoms, history, and risk factors	Multimodal fusion layer with temporal alignment	Missing, delayed, or contradictory data remain possible
Healthcare twins require traceability	Preserve source, timestamp, and value type	Mark values as measured, calculated, estimated, predicted, or simulated	Traceability improves auditability but does not guarantee clinical correctness
Clinical services require supervision	Keep healthcare professionals in the decision loop	Alerts, reports, risk summaries, and scenarios without autonomous treatment	Full clinical validation and regulatory assessment remain future work

6.3 State of the Art II: Cardiovascular Digital Twins

The second state of the art narrows the analysis to cardiovascular Digital Twins. This domain is used as a representative clinical deep-dive because it combines electrical signals, anatomical imaging, hemodynamic measurements, clinical history, longitudinal monitoring, and patient-specific physiological modeling. Figure 6.3 summarizes the main categories identified in the reviewed cardiovascular literature.

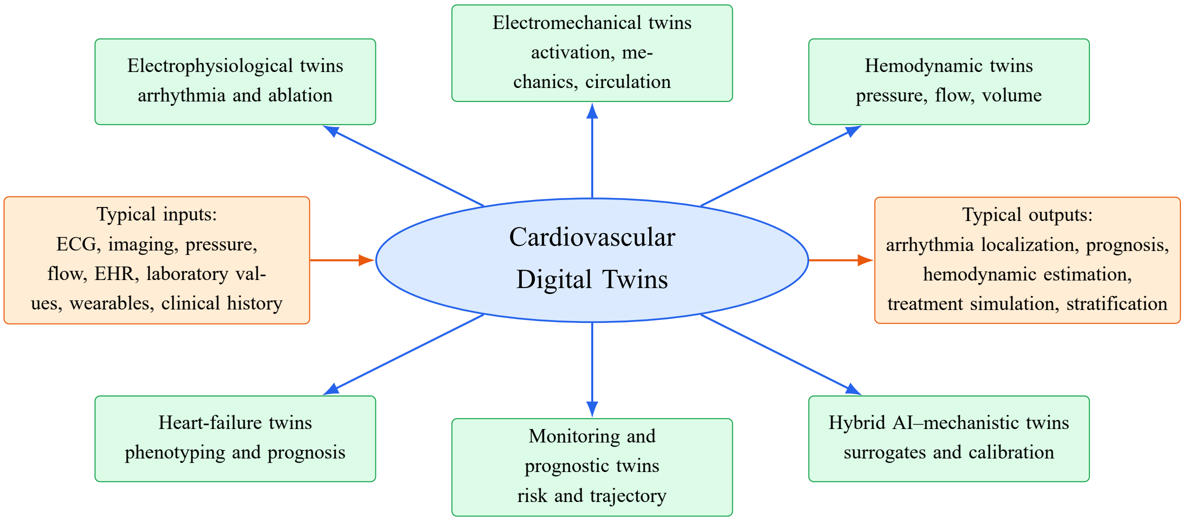

Figure 6.3: Main categories of cardiovascular Digital Twins identified in the state of the art. Author synthesis based on the reviewed literature.

6.3.1 From Precision Cardiology to Cardiac Digital Twins

Cardiovascular medicine is a representative Digital Twin domain because cardiac function emerges from coupled electrical, mechanical, fluid, anatomical, and autonomic processes. Clinical data are correspondingly multimodal: ECG describes surface electrical activity, imaging reveals anatomy and motion, catheterization measures pressures and flows, laboratory tests characterize biological state, and wearables provide longitudinal rhythm and activity data. The same diagnosis may arise from different mechanisms, which creates a strong rationale for patient-specific modeling.

Corral-Acero et al. [32] formulate the Digital Twin as an enabling concept for precision cardiology. They organize the field around statistical, mechanistic, and hybrid models and explain how patient-specific representations could support diagnosis, prognosis, device selection, and treatment simulation. Their paper is foundational because it links clinical data streams to multiscale cardiovascular models. It is also primarily a vision and agenda: many examples available at the time were personalized simulations rather than continuously synchronized twins.

Coorey et al. [33] conduct a mapping review spanning academic publications, commercial activity, and patents. They identify cardiovascular research as a major component of health Digital Twin activity but conclude that most reported applications are numerical simulation models or precursors of the real-time cyber-physical system implied by a strict Digital Twin definition. The review emphasizes precise phenotyping, multimodal integration, prediction, and treatment optimization, while identifying ethical constraints and clinical adoption barriers.

Sel et al. [34] examine the path from cardiovascular Digital Twin principles to clinical impact. Their analysis highlights multimodal sensing, mathematical models, patient-specific calibration, machine learning, and uncertainty. The paper is particularly useful for distinguishing a persistent individual representation from a one-time personalized model and for stressing that trustworthy cardiovascular twins require bias control, validation, and interpretable estimates.

Cluitmans et al. [35] focus on cardiac electrophysiology. They review applications in sudden-death risk, atrial fibrillation, ventricular arrhythmia, and rhythm-control therapy, while emphasizing the technical difficulty of calibration and the challenge of maintaining a functionally equivalent twin as the patient's condition changes. Their review shows that electrophysiology is among the most developed cardiac-twinning areas but still depends heavily on offline imaging, inverse problems, and computational modeling.

Bhagirath et al. [36] provide a complementary state-of-the-future review centered on cardiac electrophysiology. They distinguish anatomical personalization from mechanistic and functionally equivalent twinning, examine calibration and data-assimilation requirements, and review applications in pacing, cardiac resynchronization, atrial fibrillation, and ventricular tachycardia. Their analysis is especially valuable because it warns against treating every personalized model as a complete twin: the physiological attributes reproduced, the parameters calibrated, and the unmodeled aspects of the patient must be stated explicitly. The review also identifies continuous updating, automated model-generation pipelines, multimodal integration, wearable data, and credible clinical trials as prerequisites for translation.

6.3.2 Patient-Specific Electrophysiology and Arrhythmia Risk

Several influential studies predate widespread use of the Digital Twin label but provide essential technical foundations. Arevalo et al. [37] construct personalized three-dimensional post-infarction heart models from MRI and simulate arrhythmia inducibility. Their proof-of-concept study demonstrates that mechanistic patient-specific modeling can improve risk stratification beyond conventional clinical markers. The approach is an important precursor to cardiac Digital Twins, although it is an offline virtual-heart test rather than a continuously updated twin.

Prakosa et al. [38] extend personalized virtual-heart modeling to infarct-related ventricular tachycardia ablation. The study includes retrospective animal and human analyses and a small prospective feasibility component. Patient-specific imaging is converted into electrophysiological models used to identify potential ablation targets. The work provides stronger clinical translation than a purely retrospective simulation, but the modeling pipeline remains computationally intensive and is not a persistent longitudinal twin.

Zhang et al. [39] develop genotype-specific heart Digital Twins for arrhythmogenic right ventricular cardiomyopathy. The framework integrates contrast-enhanced MRI-derived structural remodeling with genotype-specific cellular electrophysiology. In 16 retrospectively studied patients, the twins predicted ventricular-tachycardia circuit locations with high reported sensitivity, specificity, and accuracy relative to clinical electrophysiology studies. The contribution is important because genotype information is incorporated into patient-specific mechanistic models. The cohort is nevertheless small, and prospective multi-center validation is needed.

Hwang et al. [40] construct patient-specific left-atrial models from computed tomography and electroanatomical maps to perform a virtual amiodarone test after atrial-fibrillation ablation. Virtual response to drug concentrations was associated with one-year recurrence outcomes, with lower recurrence in the group whose virtual AF terminated at therapeutic doses. This study illustrates a clinically meaningful treatment-simulation service and links virtual predictions to follow-up outcomes. Its limitations include retrospective design, disease- and drug-specific assumptions, and the need for external prospective replication.

O'Hara et al. [41] investigate how cardiac magnetic-resonance resolution affects Digital-Twin predictions of post-infarction ventricular-tachycardia circuits. For six patients undergoing ablation, the authors reconstructed three twins per patient from clinical-resolution and over-sampled late-gadolinium-enhanced MRI and compared predicted circuits with intraprocedural electroanatomical mapping. Higher-resolution twins reduced partial-volume effects and identified a larger proportion of unique circuits, including an emergent circuit after an unsuccessful initial ablation. The study demonstrates that imaging resolution is not a neutral preprocessing choice: it changes scar geometry, predicted re-entry pathways, and potential treatment targets. Its small cohort and preprocedural offline workflow prevent conclusions about general clinical effectiveness, but the work provides concrete validation against procedural data.

6.3.3 Whole-Heart, Electromechanical, and Hemodynamic Models

Gerach et al. [42] present a fully coupled multiscale whole-heart framework integrating electrophysiology, excitation–contraction coupling, mechanics, and closed-loop circulation. The study provides a detailed mechanistic foundation and discusses adaptation to patient-specific measurements. Its strength is physiological coverage; its limitation is computational and calibration complexity. A high-fidelity whole-heart model can support explanation and scenario testing, but without routine updating from an individual patient it remains a platform for building twins rather than a deployed twin itself.

Viola et al. [43] address computational feasibility using GPU acceleration. Their multiphysics model reproduces a heartbeat within hours and is demonstrated for left bundle branch block and cardiac resynchronization. This creates opportunities for synthetic cohorts and repeated simulation campaigns that would be impractical with slower solvers. However, simulation speed alone does not ensure patient-specific validity, and the proof-of-concept is not evidence of real-time clinical synchronization.

Salvador et al. [44] address the same computational bottleneck through a physics-informed AI surrogate. Their Latent Neural Ordinary Differential Equation model learns pressure–volume dynamics from 400 high-fidelity electromechanical simulations spanning 43 parameters related to electrophysiology, mechanics, and cardiovascular hemodynamics. The surrogate enables global sensitivity analysis, parameter estimation, and uncertainty quantification on a single processor within a few hours, illustrating how learned reduced-order models can make repeated whole-heart analyses more tractable. The contribution is methodological rather than a deployed clinical twin: training data originate from simulations of a heart-failure case, and robustness beyond the modeled parameter domain requires further patient-level validation.

Geddes et al. [45] develop patient-specific pulmonary-artery models for heart-failure monitoring. Computed-tomography pulmonary angiography is used to reconstruct anatomy, while invasive hemodynamic measurements provide boundary information and validation targets. The study investigates the minimum geometric complexity needed to predict pulmonary artery pressure and validates predictions against implantable-monitor measurements in a proof-of-concept cohort. The work is clinically relevant because it aims to derive predictive hemodynamic markers noninvasively. Its limitations include dependence on imaging quality, boundary-condition assumptions, computational workflow, and a small validation cohort.

6.3.4 AI-Enhanced Cardiovascular Twins and Population-Scale Modeling

Martinez-Velazquez et al. [46] propose Cardio Twin, an edge-oriented architecture for ischemic-heart-disease detection. The work is an early example of combining connected data, classification, and edge processing under the Digital Twin label. It demonstrates that resource-constrained devices can support parts of a cardiac-twin pipeline, but its virtual heart representation and physiological simulation are limited compared with later mechanistic frameworks.

Gu et al. [47] create patient-specific mechanistic cardiovascular models for 343 heart-failure patients. The optimized parameters and corresponding simulations are treated as Digital Twin features and used in unsupervised phenotyping and supervised prognosis. The integration of physiological prior knowledge with AI improves interpretability and transferability compared with models using clinical variables alone. The study is a strong example of model-informed AI, but measurements were assembled retrospectively, sometimes within a multi-month window, and the twins were not continuously synchronized in real time.

Qian et al. [48] demonstrate population-scale cardiac twinning by constructing 3,461 cardiac Digital Twins from UK Biobank data and 359 additional twins from an ischemic-heart-disease cohort. Cardiac MRI and ECG are used to infer electrophysiological properties, revealing associations between conduction, repolarization, anatomy, age, obesity, lifestyle, and outcomes. The scale is a major contribution because traditional high-fidelity personalization is usually limited to small cohorts. However, the resulting twins are derived from episodic research data and are not persistent clinical replicas that evolve continuously with each participant.

Recent AI surrogate approaches aim to reduce the cost of whole-heart models by learning approximations to high-fidelity simulations. These methods can enable rapid parameter estimation, sensitivity analysis, and near-real-time scenario evaluation. Their clinical credibility depends on domain-of-validity analysis, uncertainty quantification, and safeguards against extrapolation. The hybrid strategy is promising because the mechanistic model defines physiological structure while AI accelerates inference, but it also introduces two interacting error sources: the mechanistic assumptions and the learned surrogate approximation.

6.3.5 Comparative Table for Cardiovascular Digital Twins

Table 6.3 compares the principal cardiovascular studies according to clinical objective, data, modeling, evidence, and maturity. Foundational personalized-heart studies are included and explicitly labeled as precursors where they do not implement persistent synchronization. The comparison distinguishes electrophysiological models from hemodynamic and electromechanical models, high-fidelity simulations from fast surrogates, patient-specific calibration from longitudinal synchronization, and mechanistic interpretation from AI-assisted prognosis.

Table 6.3: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies.

Study	Clinical objective / entity	Patient data	Modeling approach	Maturity	Evidence / main contribution	Main limitations
Corral-Acero et al. [32]	Precision cardiology across diagnosis, prognosis, and treatment	Imaging, biosignals, clinical, molecular, and device data	Statistical, mechanistic, and hybrid modeling roadmap	Field vision	Establishes the multiscale precision-cardiology DT agenda and application taxonomy	Position paper; limited direct implementation and clinical outcome evidence
Coorey et al. [33]	Mapping cardiovascular DT research, industry, and patents	Clinical, imaging, molecular, device, and contextual data	Mapping review	Emerging field synthesis	Identifies academic, commercial, and patent activity and clarifies potential CVD applications	Most studies are simulations or precursor models rather than synchronized clinical twins
Sel et al. [34]	Principles and clinical impact of cardiovascular twins	Multimodal sensors, imaging, clinical variables, and physiological signals	Review of mathematical, AI, sensing, and validation methods	Translational framework	Connects personalization, persistent representation, uncertainty, and clinical impact	Implementation evidence remains fragmented; bias, VVUQ, updating, and workflow integration are open
Cluitmans et al. [35]	Cardiac electrophysiology, arrhythmia risk, and rhythm-control therapy	ECG, imaging, electroanatomical and electrophysiological data	State-of-the-art review of mechanistic EP twins	Domain review	Synthesizes sudden-death, AF, and VT applications and calibration challenges	Routine clinical deployment and dynamic longitudinal updating remain limited
Bhagirath et al. [36]	Mechanistic Digital Twins for cardiac electrophysiology and precision treatment	MRI, CT, ECG, EHR, smartwatch, and electrophysiological data	State-of-the-future review of anatomical, mechanistic, and functionally equivalent twins	Domain review / research agenda	Defines personalization levels and analyzes calibration, pacing, CRT, AF, VT, and translation requirements	Clinical success stories and continuously updated functionally equivalent twins remain rare
Arevalo et al. [37]	Sudden-death risk after myocardial infarction	Patient MRI and clinical outcomes	Personalized 3D electrophysiological heart models and virtual arrhythmia induction	Offline patient-specific model / DT precursor	Proof-of-concept risk stratification outperformed several conventional markers	Retrospective cohort; no continuous synchronization; computational and modeling assumptions
Prakosa et al. [38]	Guidance of infarct-related VT ablation	Cardiac imaging and electrophysiological information	Personalized virtual-heart simulation of arrhythmogenic substrate	Patient-specific intervention model	Retrospective animal/human analyses plus small prospective feasibility component	Small prospective sample; complex pipeline; not a persistent longitudinal twin
Martinez-Velazquez et al. [46]	Edge-based ischemic-heart-disease detection	Sensor, mobile, contextual, and medical-record data	Edge architecture and classification	Prototype / partial DT	Demonstrates feasibility of an edge-oriented Cardio Twin architecture	Limited physiological fidelity, personalization, and clinical validation

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Table 6.3: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies (continued).

Study	**Clinical objective / entity**	**Patient data**	**Modeling approach**	**Maturity**	**Evidence / main contribution**	**Main limitations**
Gerach et al. [42]	Whole-heart electrical, mechanical, and circulatory behavior	Designed for adaptation to ECG, pressure, volume, and imaging data	Fully coupled multiscale multiphysics model	Mechanistic DT platform	Integrates electrophysiology, mechanics, contraction, and closed-loop circulation	High computational and calibration burden; limited direct patient cohort validation
Viola et al. [43]	Whole-heart multiphysics simulation and CRT scenario	Model parameters and clinical-like cardiac behavior	GPU-accelerated cardiovascular simulation	High-fidelity simulation platform	Demonstrates LBBB and resynchronization proof of concept with accelerated computation	Hours per heartbeat are not real time; patient-specific updating and prospective validation remain absent
Salvador et al. [44]	Fast whole-heart electromechanical analysis for heart failure	Simulation-derived pressure–volume traces across 43 electromechanical and hemodynamic parameters	Latent Neural ODE surrogate trained on 400 high-fidelity simulations	AI-accelerated mechanistic DT method	Enables sensitivity analysis, parameter estimation, and uncertainty quantification on a single processor	Demonstrated on simulation-derived data for one disease setting; external patient validation and domain-of-validity controls needed
Zhang et al. [39]	VT circuit prediction in arrhythmogenic right ventricular cardiomyopathy	Contrast-enhanced MRI, genotype, and clinical EP study data	Genotype-specific structural and cellular EP heart twins	Retrospective patient-specific DT	In 16 patients, reported high agreement with clinically identified VT circuit locations	Small single-center retrospective cohort; requires prospective multi-center confirmation
Hwang et al. [40]	Prediction of amiodarone efficacy after AF ablation	CT, electroanatomical maps, fibrosis, and one-year follow-up	Patient-specific left-atrial electrophysiology and virtual drug concentrations	Treatment-simulation DT	Virtual drug response stratified patients with significantly different AF recurrence outcomes	Retrospective, disease- and drug-specific; external and prospective validation needed
O'Hara et al. [41]	Preprocedural identification of post-infarction VT circuits and ablation targets	Clinical and oversampled late-gadolinium-enhanced CMR plus intraprocedural electroanatomical mapping	Patient-specific infarct geometry and rapid-pacing electrophysiological simulation	Validated offline heart twin	In six patients, higher-resolution twins better identified unique VT circuits and emergent VT than clinical-resolution twins	Very small cohort; offline workflow; imaging-resolution dependence and uncertain generalizability
Geddes et al. [45]	Noninvasive prediction of pulmonary artery pressure in heart failure	CT pulmonary angiography, catheterization, and implantable monitor data	Patient-specific geometry and 3D CFD	Calibrated proof-of-concept DT	Evaluates geometry complexity and validates pressure prediction against measured data	Small cohort; boundary conditions, imaging, and computation limit scalability
Gu et al. [47]	Heart-failure phenotyping, diagnosis, and prognosis	EHR, TTE, CMR, RHC, pressure, flow, and volume data	Mechanistic cardiovascular models plus unsupervised and supervised AI	Retrospective patient-specific model / intelligent-DT direction	343 patient twins; physiologically interpretable features improved phenotyping and prognostic modeling	Retrospective data and measurement-time mismatch; not continuously synchronized; prospective validation required

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Table 6.3: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies (continued).

Study	Clinical objective / entity	Patient data	Modeling approach	Maturity	Evidence / main contribution	Main limitations
Qian et al. [48]	Population-scale electro-physiological mechanism discovery	Cardiac MRI, ECG, lifestyle, functional, and outcome variables	Physics-constrained cardiac twins with machine-learning-enabled personalization	Large-scale research cohort twins	3,461 UK Biobank and 359 IHD twins reveal mechanistic associations and validate selected findings	Episodic research data rather than live clinical twins; inferred parameters depend on model assumptions

6.3.6 Synthesis of the Cardiovascular State of the Art

The cardiovascular literature is more technically mature than many other eHealth domains in terms of mechanistic modeling, patient-specific geometry, and simulation of clinically meaningful interventions. It includes prospective feasibility work, treatment-response studies, large-scale cohorts, and validation against invasive measurements. Nevertheless, most systems remain calibrated snapshots rather than persistent twins. They are commonly reconstructed from imaging and episodic clinical measurements, analyzed offline, and not continuously updated as disease, treatment, and behavior change.

Three modeling directions can be distinguished. Electrophysiological twins simulate activation, repolarization, arrhythmia induction, drug effects, and ablation. Electromechanical and hemodynamic twins estimate pressures, flows, volumes, tissue mechanics, and device response. Hybrid and AI-enhanced twins use mechanistic parameters as interpretable features or train surrogates that accelerate expensive simulations. The strongest emerging systems combine these directions, but each additional modality increases calibration, identifiability, uncertainty, and validation challenges.

A recurring technical problem is non-uniqueness: different parameter combinations can reproduce similar clinical observations. A twin can therefore fit an ECG or pressure curve while representing the wrong underlying physiology. Credible cardiovascular twins need uncertainty distributions, sensitivity analysis, independent measurements, and validation against outcomes not used for calibration. This is especially important when the twin is used to recommend an ablation target, predict treatment response, or estimate an invasive marker.

The cardiovascular table also reveals a tension between physiological richness and deployability. Electrophysiological and electromechanical twins can explain mechanisms and simulate interventions, but they often require imaging, expert segmentation, specialized calibration, and high computational effort. AI-enhanced and population-scale approaches improve speed and scalability, yet they may depend on retrospective cohorts, inferred parameters, or simulation-derived training data. For this reason, the most credible direction is not to replace mechanistic modeling with AI, but to combine mechanistic constraints, data-driven estimation, uncertainty reporting, and clinically realistic validation.

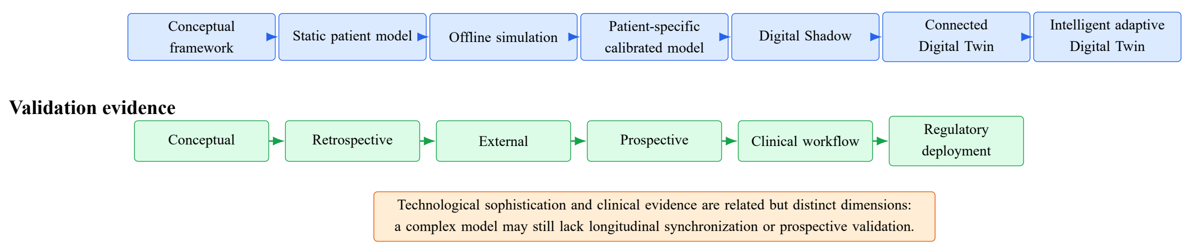

Figure 6.4: Evidence and maturity continuum for healthcare and cardiovascular Digital Twins. Author synthesis based on the reviewed literature.

6.3.7 Implications of the Cardiovascular State of the Art for CardioTwin

The cardiovascular literature informs the practical scope of CardioTwin in two ways. First, it demonstrates why the cardiovascular domain is a suitable specialization: cardiac assessment combines electrical activity, hemodynamic variables, clinical symptoms, risk factors, imaging, and longitudinal follow-up. Second, it shows the limits of what can be claimed in an academic prototype. High-fidelity cardiac twins may require three-dimensional anatomy, invasive pressure measurements, patient-specific boundary conditions, electrophysiological calibration, or expensive multiphysics computation. CardioTwin is therefore positioned as a lower-complexity patient-state cardiovascular twin, not as a complete virtual heart.

In this scope, ECG provides important electrical information but cannot alone describe the full cardiovascular state. SBP, DBP, heart rate, clinical history, symptoms, and risk factors provide complementary information for risk assessment and longitudinal monitoring. Mechanistic cardiovascular studies justify the need for physiological interpretation, while AI supports ECG analysis, pattern recognition, patient stratification, and risk estimation. What-if scenarios must remain evidence-based, transparent, and uncertainty-aware: population-level medication or lifestyle effects cannot be interpreted as guaranteed individual outcomes.

CardioTwin is therefore inspired by the reviewed cardiovascular Digital Twin literature but does not claim to implement full three-dimensional cardiac anatomy, complete electromechanical simulation, invasive pulmonary-pressure modeling, autonomous treatment optimization, or clinically certified personalized drug-response prediction. These functions remain outside the PFE scope and would require specialized datasets, clinical protocols, regulatory consideration, and prospective validation.

Table 6.4: Cardiovascular state-of-the-art findings and their translation into the CardioTwin PFE.

Cardiovascular DT finding	Relevance to CardioTwin	CardioTwin implementation	Out-of-scope element
ECG is one modality among several	Avoid an ECG-only representation of the patient	ECG analysis combined with BP, symptoms, history, and risk factors	Complete cardiac anatomy and full organ reconstruction
High-fidelity models require calibration	Use explicit patient-state variables and transparent assumptions	Patient-specific measurements, history, and update rules	Full CFD or electromechanical calibration
Hybrid modeling improves interpretation	Combine AI outputs with clinical reasoning and scenario logic	ECG AI, clinical assessment, risk estimation, and controlled what-if rules	Autonomous therapy optimization
Longitudinal data support personalization	Preserve patient history and repeated measurements	State storage, trend visualization, and repeated updates	Continuous hospital-grade monitoring
Uncertainty is unavoidable	Display assumptions, warnings, and limits of interpretation	Transparent scenario outputs and human-supervised decisions	Guaranteed individual treatment response

6.4 Cross-Domain Comparison and Research Gaps

The two states of the art reveal a gap between architectural completeness and physiological fidelity. General eHealth research provides layered architectures, interoperability principles, governance requirements, and broad service taxonomies. Cardiovascular research provides detailed mechanistic and patient-specific models with stronger physiological interpretation. Few studies combine both strengths in a single system that is interoperable, longitudinal, dynamically updated, clinically validated, and usable in routine care.

6.4.1 General eHealth Digital Twin Gaps

The general healthcare literature identifies several unresolved problems that are not limited to one clinical specialty:

- **Definitions and benchmarks:** Digital Model, Digital Shadow, partial connected Digital Twin, process Digital Twin, and intelligent Digital Twin are not always distinguished consistently despite the definition framework discussed in Chapter 3.

- **Synchronization:** many systems remain one-way dashboards or episodic simulations rather than continuously maintained physical–virtual systems.
- **Data fragmentation:** EHR data, imaging, laboratory results, wearables, and patient-reported data remain difficult to integrate across institutions and time scales.
- **Interoperability and provenance:** robust semantic mapping, source traceability, units, timestamps, and data-quality indicators are still weak in many prototypes.
- **Longitudinal representation:** few systems demonstrate stable patient-state maintenance across repeated measurements, treatments, and disease progression.
- **Clinical validation:** external, prospective, multi-center, workflow-level, and outcome-oriented validation remain limited.
- **Governance and responsibility:** privacy, consent, accountability, explainability, cybersecurity, and clinical responsibility remain unresolved implementation barriers.

6.4.2 AI-Related Gaps

AI increases the analytical power of Digital Twins but also creates specific gaps. Complex models require explainability, calibration, subgroup analysis, bias assessment, uncertainty estimation, and out-of-distribution testing. Adaptive models require drift detection and documented revalidation. Generative AI and LLM-based components require governance because synthetic outputs may reproduce bias, leak private information, or appear more clinically reliable than they are. A major methodological gap is the limited comparison between pure AI systems, pure mechanistic simulations, and hybrid approaches under the same clinical objectives. Another gap is the risk of interpreting overfitting as personalization when the model has not been externally validated.

6.4.3 Cardiovascular Digital Twin Gaps

The cardiovascular literature adds domain-specific challenges. High-fidelity models often depend on specialized imaging, invasive measurements, patient-specific segmentation, boundary-condition estimation, and expensive computation. Physiological parameters may be difficult to identify uniquely, and several parameter sets can explain the same ECG, pressure, or flow observations. Longitudinal synchronization remains uncommon, multicenter validation is still limited, and many studies treat ECG, imaging, pressure, flow, and clinical history separately rather than as a continuously updated patient-state representation. What-if treatment simulation remains scientifically promising but clinically sensitive because predicted medication or intervention effects require prospective validation before individual therapeutic use.

6.4.4 CardioTwin Scope

CardioTwin addresses a limited and realistic subset of these gaps. It focuses on multimodal cardiovascular patient-state representation, real ECG and BP acquisition, AI-assisted ECG analysis, integration of clinical and physiological variables, longitudinal state updates, risk-oriented summaries, evidence-based what-if scenarios, transparent uncertainty warnings, and clinician-facing visualization. It does not claim to solve full organ-scale cardiac simulation, continuous hospital-grade synchronization, invasive hemodynamic personalization, autonomous treatment optimization, regulatory certification, or prospective clinical validation. These limits are necessary to keep the PFE scientifically credible while still making it a practical continuation of the Master's thesis findings.

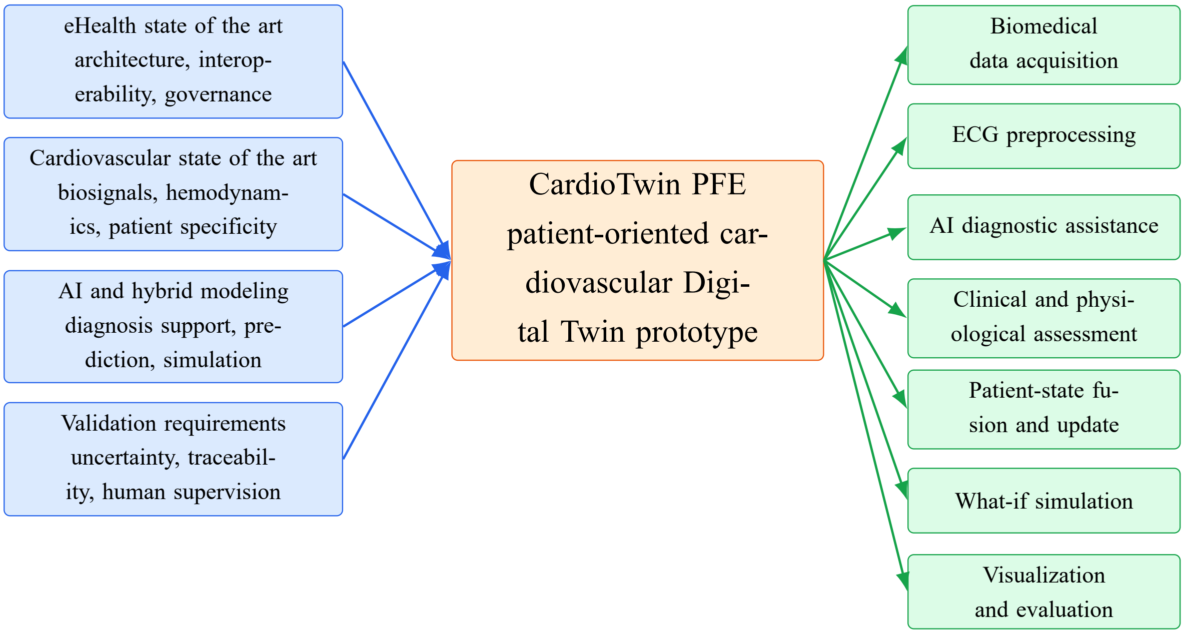

Figure 6.5: Translation of the Master's thesis state-of-the-art findings into the CardioTwin PFE methodology. Author synthesis based on the reviewed literature.

A future clinically credible eHealth Digital Twin should therefore be defined not only by model sophistication but by a complete evidence chain: verified software, traceable and interoperable data, patient-specific calibration, uncertainty-aware prediction, external and prospective validation, human-supervised services, and controlled feedback. Cardiovascular studies offer strong examples of physiological modeling and treatment simulation, while general eHealth studies provide the architectural and governance foundation required to translate these capabilities into broader healthcare systems.

6.5 Conclusion

This chapter developed two distinct but connected states of the art. The first showed how healthcare Digital Twins extend from patient-specific medicine to hospital processes, public health, and AI-enabled services. The second examined cardiovascular Digital Twins in depth, including arrhythmia risk, ablation planning, virtual drug testing, whole-heart modeling, heart-failure hemodynamics, interpretable AI, and population-scale twinning. The comparative analyses demonstrate that the field has progressed beyond purely conceptual discussion, but fully connected and clinically validated patient twins remain uncommon. The chapter also translated the literature findings into explicit CardioTwin design implications, establishing the scientific basis for the proposed method and final PFE transition.

Part IV

Proposed Method

CHAPTER 7

PROPOSED ARCHITECTURE FOR A DIGITAL TWIN-BASED EHEALTH SYSTEM

7.1 Architecture Rationale

Based on the literature review, a Digital Twin-based eHealth system should not be limited to a data dashboard or a predictive algorithm. It should be organized as a layered system that connects the physical healthcare environment to a virtual representation, processes heterogeneous data, provides simulation and analysis services, and ensures security, privacy, and interoperability.

The proposed architecture is general and can be adapted to different eHealth scenarios such as remote monitoring, chronic disease management, hospital workflow optimization, rehabilitation support, wellness tracking, or clinical decision support. It is not designed for one specific disease or one specific project.

7.2 Overview of the Proposed Architecture

The proposed architecture is composed of seven layers: physical layer, data acquisition layer, interoperability and preprocessing layer, data management layer, virtual twin layer, intelligence and simulation layer, and service layer. Security, privacy, and governance are considered as cross-cutting concerns across all layers.

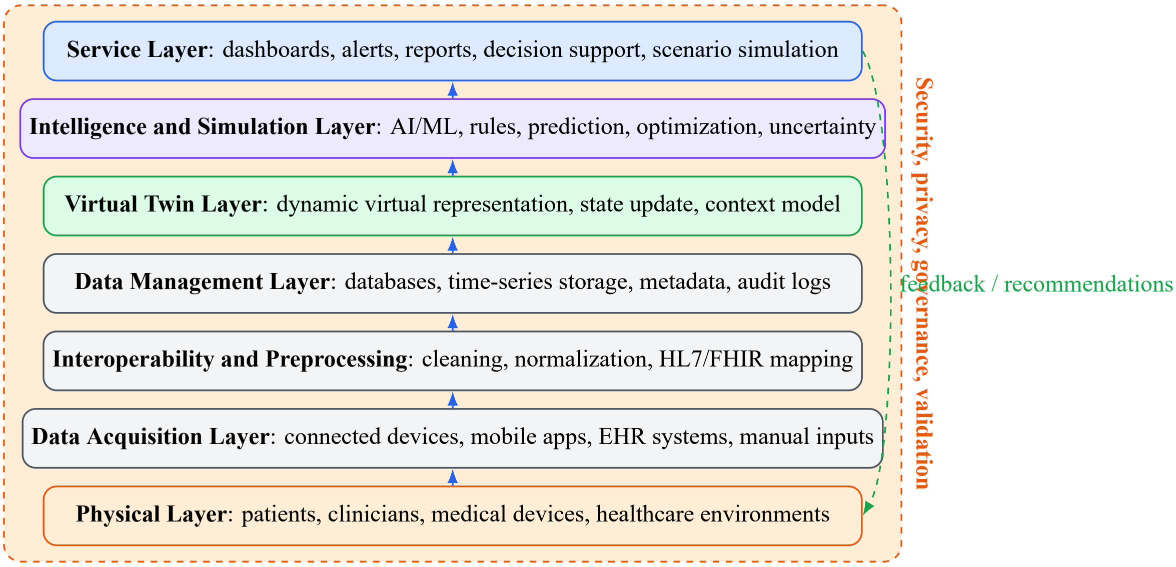

Figure 7.1: Proposed multi-layer architecture for a Digital Twin-based eHealth system. Author synthesis based on the reviewed literature.

7.3 Physical Layer

The physical layer includes real-world entities involved in the eHealth ecosystem. These may include patients, healthcare professionals, medical devices, wearable sensors, mobile phones, hospital equipment, and healthcare environments. The role of this layer is to generate real-world data and receive possible feedback or recommendations from the digital services.

7.4 Data Acquisition Layer

The data acquisition layer collects information from connected devices, mobile applications, clinical systems, and manual inputs. Data may be continuous, periodic, or event-based. Examples include physiological measurements, activity data, medication information, symptoms, clinical notes, laboratory results, and operational data.

7.5 Interoperability and Preprocessing Layer

This layer prepares data for use by the Digital Twin. It handles data cleaning, normalization, timestamp alignment, missing value management, anomaly detection, and format conversion. It also supports interoperability standards such as HL7 and FHIR to connect the Digital Twin with healthcare information systems.

7.6 Data Management Layer

The data management layer stores historical, real-time, and contextual information. It may include relational databases, time-series databases, clinical repositories, data warehouses, and metadata catalogs. Data governance rules must define who can access data, how long data are stored, and how data are anonymized or pseudonymized when needed.

7.7 Virtual Twin Layer

The virtual twin layer represents the current state of the physical entity or healthcare process. Depending on the application, the virtual twin may represent a patient profile, a disease trajectory, a medical device, a care pathway, a hospital service, or a public health process. The virtual twin should be continuously updated when new valid data become available.

7.8 Intelligence and Simulation Layer

This layer provides analytical and predictive capabilities. It may include rule-based reasoning, machine learning models, deep learning models, statistical models, mechanistic simulations, optimization algorithms, and uncertainty estimation. The purpose is to transform the virtual representation into useful knowledge for monitoring, prediction, scenario analysis, and decision support.

7.9 Service Layer

The service layer exposes the functions of the Digital Twin to users and systems. Possible services include dashboards, alerts, reports, risk assessment, scenario simulation, remote monitoring, workflow optimization, and decision support. These services must be designed according to user needs, clinical safety, explainability, and usability requirements.

7.10 Security, Privacy, and Governance Layer

Security and privacy are cross-cutting concerns. The system should include authentication, authorization, encryption, audit logs, secure APIs, role-based access control, data minimization, consent management, and governance policies. Ethical oversight is also required to ensure fairness, transparency, and responsible use of the Digital Twin.

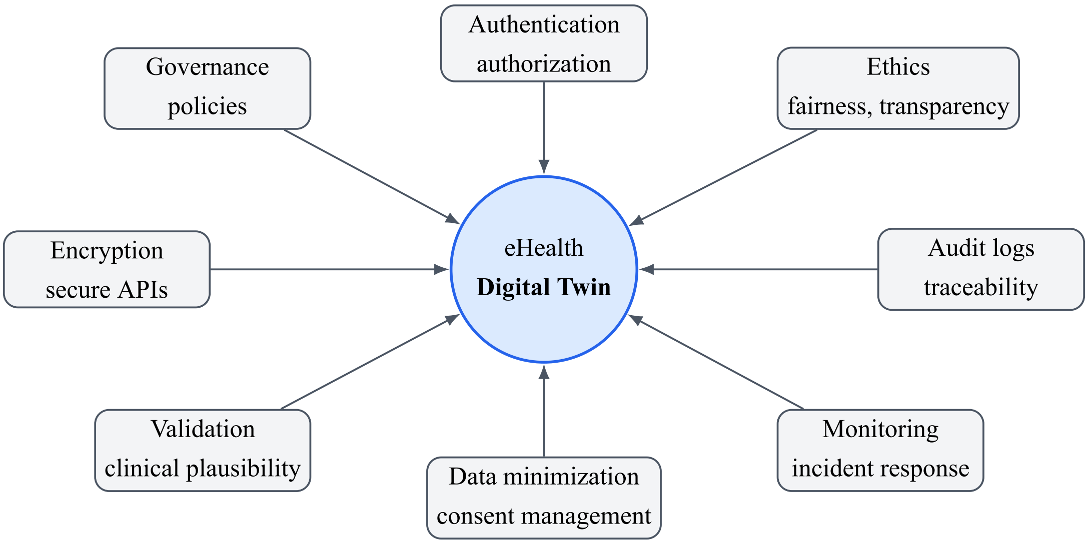

Figure 7.2: Security, privacy, and governance concerns around the eHealth Digital Twin. Author synthesis based on the reviewed literature.

7.11 Data Flow

Figure 7.3 illustrates the general data flow. Real-world data are collected from the physical layer, transmitted through acquisition modules, preprocessed and standardized, stored in the data layer, and used to update the virtual twin. The intelligence and simulation layer generates insights that are exposed through services. Feedback may then be provided to patients, clinicians, or healthcare systems depending on the use case.

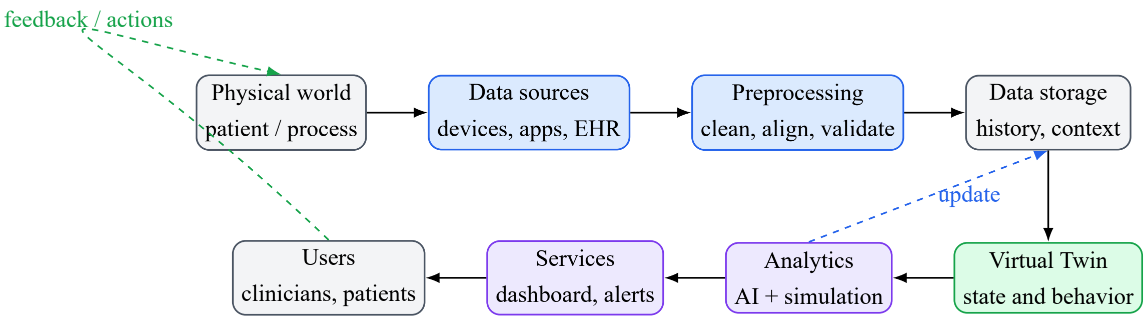

Figure 7.3: General data flow in a Digital Twin-based eHealth system. Author synthesis based on the reviewed literature.

7.12 General Use Case Scenario

A general eHealth use case can be described as follows. A patient uses connected devices and a mobile application to collect health-related data at home. These data are transmitted to the eHealth platform, cleaned, standardized, and stored. The virtual twin is updated with the new measurements and historical context. The intelligence layer analyzes trends, detects anomalies, estimates risk, and simulates possible scenarios. The service layer displays a dashboard, generates alerts when necessary, and produces a report for healthcare professionals.

This scenario remains general and can be adapted to different domains such as chronic disease follow-up, elderly care, rehabilitation, mental health monitoring, wellness tracking, hospital-at-home programs, and clinical workflow optimization.

7.13 Evaluation Plan

The proposed architecture can be evaluated using several criteria:

- **Functional evaluation:** ability to collect, store, update, analyze, and display data.
- **Interoperability evaluation:** compatibility with standards and existing healthcare systems.
- **Performance evaluation:** latency, scalability, and reliability.
- **Security evaluation:** authentication, authorization, encryption, and auditability.
- **Usability evaluation:** clarity of dashboards and usefulness for users.
- **Clinical plausibility:** relevance and safety of generated outputs.
- **Ethical evaluation:** fairness, transparency, consent, and data governance.

7.14 Conclusion

This chapter proposed a general conceptual architecture for Digital Twin-based eHealth systems. The architecture is designed to be reusable, modular, secure, interoperable, and adaptable to multiple healthcare scenarios.

Part V

General Conclusion and PFE Transition

CHAPTER 8

GENERAL CONCLUSION AND PROPOSED METHOD FOR THE PFE

8.1 General Conclusion

This Master's thesis studied the Digital Twin paradigm in the context of eHealth. It showed that a Digital Twin is not only a digital copy, dashboard, or isolated prediction model, but a connected physical–virtual system that uses data, models, and services to support monitoring, simulation, prediction, and decision support. The thesis also distinguished digital models, digital shadows, connected Digital Twins, and intelligent Digital Twins in order to avoid confusing static simulations with mature, updated, and clinically meaningful systems.

The first part of the study clarified the foundations of eHealth, Digital Twin technology, and healthcare Digital Twin applications. The state-of-the-art analysis then examined two complementary domains. The first state of the art reviewed Digital Twins in eHealth and healthcare, including patient-specific twins, organ and device twins, hospital-process twins, public-health twins, Artificial Intelligence-enabled systems, hybrid AI–simulation models, interoperability, ethics, privacy, governance, and validation. The second state of the art focused on cardiovascular Digital Twins, including electrophysiology, arrhythmia modeling, ablation planning, electromechanical and hemodynamic models, heart-failure twins, patient-specific calibration, AI-accelerated simulation, and population-scale cardiac twins.

The comparative analyses confirmed that the field is scientifically active but not yet uniformly mature. Many studies remain conceptual, retrospective, or proof-of-concept. Fully synchronized, longitudinal, clinically validated, and workflow-integrated healthcare Digital Twins remain uncommon. The thesis therefore emphasized cautious terminology, traceable data flows, uncertainty-aware interpretation, human supervision, and the distinction between measured, calculated, estimated, predicted, and simulated values.

8.2 Main Contributions of the Master Study

The main contributions of this work are:

- a structured review of eHealth and Digital Twin foundations;
- a dual state-of-the-art study covering general healthcare Digital Twins and cardiovascular Digital Twins;
- two landscape comparative tables that compare studies by represented entity, data, modeling approach, maturity, evidence, and limitations;
- synthesis figures that clarify AI roles, eHealth Digital Twin categories, cardiovascular Digital Twin categories, maturity levels, and the transition toward CardioTwin;
- a general multi-layer architecture for Digital Twin-based eHealth systems;
- a literature-derived rationale for translating the Master's thesis findings into the CardioTwin PFE.

8.3 Limitations of the Study

This thesis is primarily a state-of-the-art and conceptual architecture study. It does not provide a complete clinical deployment, regulatory medical device, or prospectively validated healthcare product. The reviewed literature is also heterogeneous: authors use the term Digital Twin with different levels of rigor, and many publications describe simulations, predictive models, or digital shadows rather than fully synchronized twins.

Another limitation is that the proposed architecture remains general. It identifies layers, data flows, and cross-cutting requirements, but each healthcare use case still requires specific datasets, clinical objectives, validation protocols, regulatory analysis, and user requirements. In cardiovascular medicine, high-fidelity twinning may require imaging, invasive measurements, calibration, and computational resources that are beyond the scope of this Master's thesis and the associated academic prototype.

8.4 From the Master Study to the PFE

The CardioTwin PFE is introduced as a practical cardiovascular continuation of the Master's thesis. The transition follows a clear logic: general Digital Twin foundations define the physical–virtual relationship; eHealth foundations define the healthcare data and service environment; the healthcare state of the art identifies architectural, AI, interoperability, and governance requirements; the cardiovascular state of the art identifies a clinically relevant specialization; and the synthesis translates these findings into a feasible prototype scope.

The cardiovascular domain was selected because it satisfies several technical and methodological criteria identified in the literature. Cardiovascular diseases require longitudinal and personalized monitoring; the domain combines electrical, hemodynamic, physiological, clinical, and sometimes imaging information; patient presentations are heterogeneous; disease trajectories change over time; and treatment responses vary between patients. The reviewed literature also provides examples ranging from numerical simulation and patient-specific calibrated models to AI-assisted phenotyping and prognosis.

8.5 Introduction to the PFE Project

The PFE is entitled:

> **CardioTwin – Design and Deployment of an AI-Powered Cardiovascular Digital Twin for Diagnostic Assistance and Therapeutic Simulation**

CardioTwin is positioned as a patient-oriented cardiovascular Digital Twin prototype inspired by the literature synthesis. It does not claim to reproduce a complete three-dimensional virtual heart or to provide autonomous medical prescriptions. Instead, it aims to combine real biomedical measurements, ECG signal analysis, clinical and physiological information, patient-state representation, cardiovascular risk assessment, controlled what-if simulation, monitoring, and visualization services.

8.6 Proposed PFE Methodology

The proposed method follows the layered and evidence-aware logic derived from the Master's thesis.

8.6.1 Data Acquisition and Validation

CardioTwin will acquire or import ECG data, systolic blood pressure, diastolic blood pressure, heart rate when available, demographic information, clinical history, symptoms, and other relevant physiological measurements. The prototype may use connected medical sensors and the MySignals eHealth development kit where appropriate, but hardware acquisition remains a means to support the patient-state model rather than an objective in itself. Preprocessing includes ECG filtering, artifact handling, normalization, segmentation, timestamp alignment, missing-data handling, and validation of measurement consistency.

8.6.2 ECG and Clinical Assessment

The ECG diagnostic-assistance module analyzes biosignals for rhythm or abnormality detection using suitable Machine Learning or Deep Learning models and biomedical datasets such as PTB-XL, MIT-BIH, SVDB, AFDB, or INCART when required by the implementation. These outputs are diagnostic-support elements only. They are combined with age, sex, SBP, DBP, heart rate, clinical history, symptoms, and relevant cardiac measurements to build a broader cardiovascular assessment.

8.6.3 Patient-State Fusion and Twin Update

The virtual patient state fuses ECG findings, clinical risk, physiological measurements, and historical information. This state is more than an isolated ECG classifier because it preserves the relationship between measurements, derived indicators, model estimates, predictions, and simulated outputs. The cardiovascular twin is updated when new validated information is received, while explicitly distinguishing measured values from calculated, estimated, predicted, and simulated values.

8.6.4 What-If Simulation and Services

The what-if module is an exploratory decision-support component. It may simulate changes in blood pressure, physiological parameters, lifestyle scenarios, medication-effect assumptions, or possible evolution over a selected time horizon. These scenarios are estimates based on models, evidence, or population-level effects and must not be interpreted as guaranteed individual outcomes or autonomous medical prescriptions. The service layer provides patient dashboards, ECG visualization, historical trends, risk summaries, alerts, what-if scenario outputs, and clinician-oriented reports.

8.6.5 Human-in-the-Loop Decision Support

CardioTwin assists healthcare professionals but does not replace them. Clinical decisions remain under professional supervision, simulated recommendations require medical interpretation, and autonomous therapeutic control is outside the scope of the PFE.

8.7 Evaluation Strategy

The evaluation plan covers several complementary levels. Data acquisition evaluation verifies sensor communication, completeness, timestamps, signal quality, and measurement consistency. ECG-model evaluation uses appropriate metrics such as accuracy, precision, recall, F1-score, macro-F1, sensitivity, specificity, confusion matrices, and external dataset testing where available. Patient-state evaluation examines consistency between measurements and stored state, correct longitudinal updates, and handling of missing or contradictory measurements.

What-if evaluation focuses on traceability of assumptions, consistency with cited medical evidence, plausible direction and magnitude of simulated effects, transparent uncertainty, and the absence of unsupported claims of individual treatment efficacy. System evaluation considers latency, modularity, reliability, usability, security, interoperability, and dashboard clarity. Full clinical validation would require clinician review, prospective evaluation, ethical approval where necessary, representative patient data, and regulatory consideration; these steps may remain future work beyond the academic prototype.

8.8 Future Work

Future work should first implement and evaluate the CardioTwin prototype. Important directions include integrating real biomedical sensors, improving ECG and clinical-data fusion, validating patient-state updates, strengthening what-if simulation, adding explainability and uncertainty communication, using interoperability standards, and conducting clinician-centered usability evaluation. Longer-term work should move toward prospective validation, regulatory analysis, multicenter data, and extension to other healthcare Digital Twin domains.

8.9 Final Remarks

Digital Twins for eHealth represent a demanding research direction at the intersection of healthcare, computer science, data engineering, Artificial Intelligence, and systems modeling. Their success will depend not only on model performance, but also on synchronization, evidence, transparency, interoperability, privacy, clinical relevance, and responsible human-supervised use. CardioTwin is a practical continuation of this Master's thesis because it operationalizes a focused subset of these findings in the cardiovascular domain while preserving the scientific limits identified by the state of the art.

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APPENDIX A

APPENDIX: SUPPLEMENTARY REVIEW DIMENSIONS

This appendix summarizes the completed analytical dimensions used to compare the reviewed studies. It is included as supporting material for the two state-of-the-art tables in Chapter 6.

A.1 Study Classification Dimensions

Table A.1: Supplementary classification dimensions used during the structured literature review.

Dimension	Information extracted from each study
Represented entity	Patient, organ, physiological system, device, hospital process, care pathway, public-health process, or population.
Data modalities	EHR data, biosignals, imaging, laboratory results, sensor streams, wearable data, clinical history, workflow events, or simulated data.
Modeling approach	Mechanistic model, statistical model, Machine Learning, Deep Learning, hybrid AI–mechanistic model, graph model, process simulation, or architecture framework.
Synchronization level	Static model, offline simulation, episodic calibration, digital shadow, partial Digital Twin, process Digital Twin, connected Digital Twin, or intelligent Digital Twin direction.
Services	Monitoring, diagnosis support, prognosis, treatment simulation, resource optimization, risk assessment, alerting, reporting, or clinical workflow support.
Validation evidence	Conceptual discussion, retrospective evaluation, external validation, prospective feasibility, workflow validation, clinical deployment, or regulatory consideration.
Limitations	Data quality, calibration, missing modalities, lack of prospective validation, interoperability, privacy, explainability, uncertainty, computation, or clinical adoption.

A.2 Value-Type Traceability

For CardioTwin and related healthcare Digital Twin prototypes, the type of each stored value should remain explicit. A directly measured value, such as a blood-pressure reading, does not have the same evidential status as a calculated index, a model-estimated latent state, a predicted future trajectory, or a simulated what-if output. This distinction supports auditability, uncertainty communication, and human-supervised interpretation.